Lexicographic Minimax Regret over Declared Monotone Probes for Robust Multicriteria Selection

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Abstract

Pareto efficiency is necessary but not sufficient for a unique recommendation. An efficient set tells a decision maker (DM) which alternatives are not obviously bad, but offers no principled way to choose one. Practice fills this gap with *post hoc* devices—weighted sums, TOPSIS, compromise programming, reference points, acceptability indices—each injecting implicit or stochastic preference judgements.

We propose the **Leximax Universal-Regret (LexUR) core**, a lexicographic minimax regret selection rule operating over a structured, declared, monotone scalarisation family. The DM defines an explicit family of rational questions (probes) and supplies normalisation bounds. LexUR treats each probe's evaluation as an achievement function, calculating how far each alternative falls from the best attainable outcome under each probe, and sorts the resulting dimensionless regrets. The method returns either a singleton recommendation or, when margins or bounds are insufficient, a certified stability class, always accompanied by an interpretable certificate listing the binding questions.

We prove existence and completeness, Pareto compatibility, stability of the tolerance-based leximax rule under bounded perturbations, and a probe perturbation bound with explicit constants. We also give an adaptive correlation-clustering construction empirically reducing, without eliminating, sensitivity to redundant criteria.

In a replicated benchmark and a broadened frozen audit protocol covering 11 methods, 10 held-out preference families, 8 front geometries, and m up to 20, LexUR is competitive with robust scalarisation baselines on tail held-out loss and non-inferior to ASF and sampled minimax regret under paired tests, while a single-point hypervolume score obtains the best average rank. LexUR's distinct contributions are its labelled regret certificate and empirically reduced redundancy sensitivity relative to averaging methods. Because point-bound LexUR can be sensitive to normalisation estimates, robust deployment should report a tolerance or interval stability class; a unique recommendation is returned only when margin conditions hold. The method is robust with respect to the declared probe family, not with respect to all possible preferences. We formulate an exploratory stochastic extension illustrated on a synthetic smart-grid economic dispatch model.

Keywords: Decision support, Multicriteria decision analysis, Robust decision making, Minimax regret, Achievement scalarising functions, Pareto efficiency

1. Introduction

1.1. The Pareto front as a safety filter, not a decision object

Since the work of Pareto, efficiency has been the cornerstone of multicriteria analysis: an alternative is Pareto efficient if no criterion can be improved without worsening another. This is a logically impeccable *safety property*—it removes alternatives that are dominated and therefore indefensible. It is, however, *decisionally incomplete*. For m conflicting criteria the efficient set is typically large or infinite, and the decision maker (DM) is left with a cloud of efficient alternatives and no principled rule for selecting one.

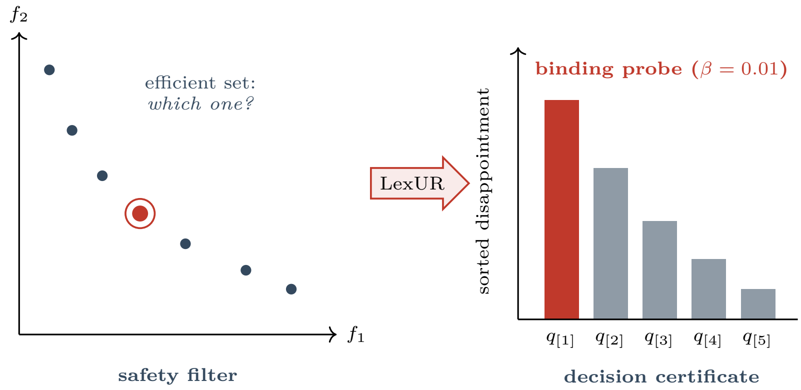

Figure 1: From a front to a certificate. The efficient set (left) removes dominated options but leaves the choice open. LexUR returns a recommendation or stability class together with an illustrative sorted regret certificate (right). A probe is called binding only when its disappointment lies within the declared presentation band $\beta = 0.01$ of the maximum; the remaining bars are secondary lexicographic constraints.

Confronted with this gap, established decision-support frameworks employ various selection devices to integrate preference information: a weighted sum requires weights [17]; TOPSIS [12] requires a distance metric and weights; compromise programming [32] requires a metric exponent and weights; outranking methods such as ELECTRE [25] and PROMETHEE [5] require concordance/discordance thresholds; and hypervolume-based selection [33] requires a reference point. These methods are robust and widely used, each with different preference assumptions. However, they require the analyst or DM to commit to additional modelling choices—weights, metrics, thresholds, exponents, or reference points—that may be difficult to justify when preference information is intentionally incomplete. The fundamental issue remains that **the Pareto front is a safety filter, not a complete decision rule**.

1.2. From a front to a certificate

We argue that, for the purpose of *choosing*, the efficient set is often insufficient as a final decision object. We propose supplementing it with a *regret certificate*: a vector of normalised disappointments over the rational questions the DM might reasonably ask about the criteria, sorted from worst to best. Rather than asking "which alternatives are efficient?" we ask

which alternative is least fragile across all declared rational interpretations of the criteria?

This turns the output of the analysis from a *set* into a *certified stability class*: a robust set of alternatives (often a singleton, if bounds are verified and margins hold) accompanied by an explanation of which questions remain most severely binding.

It is crucial to note what this approach does and does not do. LexUR is not parameter-free; preferences are represented by an explicit auditable probe family, and it requires normalisation bounds and tolerance settings for numerical implementation. Its contribution is that these choices are declared, auditable, and separated from the final recommendation, not that they disappear. Figure 1 contrasts the efficient set as a safety filter with the regret certificate as the decision object.

1.3. The Leximax Universal-Regret core

We formalise this as the **Leximax Universal-Regret (LexUR) core**.[1] At its core, LexUR is a *lexicographic minimax regret* selection rule operating over a structured, declared, monotone scalarisation family. Given a feasible set and a *declared* class $\mathcal{Q}$ of admissible monotone probes (scalar "rational

[1]The qualifier "Universal" denotes universality *over the declared probe family*—LexUR controls regret uniformly across every declared question—*not* robustness over all admissible preferences. We repeat this qualification where the distinction matters.

questions" over the normalised criteria), LexUR treats each probe as an achievement function, ranks alternatives by the lexicographically sorted vector of their normalised disappointments across $\mathcal{Q}$, and returns the lexicographic minimiser together with its certificate. The construction rests on three principles: (i) *declared admissibility*—$\mathcal{Q}$ is part of the problem statement, not hidden inside a scalarisation; (ii) *Pareto compatibility*—under monotone probes a dominated alternative is never preferred to its dominator; and (iii) *direct finite computation*—the method is primarily a *post hoc* robust selection rule operating exactly on a finite candidate set, though we also provide exact front-free formulations for structured continuous cases (§4).

1.4. Positioning and contribution

LexUR belongs to *robust* multicriteria decision analysis. Its closest relatives are stochastic multicriteria acceptability analysis (SMAA) [15, 27], which integrates acceptability over a preference distribution; robust ordinal regression [10, 9], which reasons over the set of value functions compatible with elicited holistic judgements; and preference robust optimisation [1, 11], which optimises against an ambiguity set of utility functions. Rather than replacing these methods, LexUR targets the specific setting where the analyst can declare admissible monotone questions but does not wish to elicit precise weights, thresholds, or a probability distribution over preferences. LexUR differs in object and output: it is a complementary finite-candidate robust selector that computes the *lexicographic minimax* of normalised achievement shortfalls over an explicit, monotone probe family, returning an interpretable certificate. We show that achievement scalarisation [30], the weighted sum, and the lexicographic maximin [21] are recovered as special cases of the probe family. Ultimately, LexUR provides a concrete synthesis of these established aggregation mechanisms into a single, auditable robust selection rule equipped with a linear-size probe family.

Our contributions are:

- **Core theoretical contribution:** We define the LexUR order and its auditable certificate, proving existence, completeness, Pareto compatibility *under an explicit separation condition*, dominated-solution exclusion, and stability of the tolerance-based rule under bounded perturbations (§3).
- **Computational contribution:** We present an exact candidate-set algorithm based on sequential lexicographic sorts, and outline a conditional formulation for structured continuous cases (§4).
- **Probe construction:** We propose an adaptive, redundancy-aware correlation-clustering probe family with empirical redundancy tests and a perturbation bound (§5).
- **Empirical validation:** We validate LexUR on a broadened benchmark suite with held-out additive and non-additive preferences, and apply it to a stochastic smart-grid dispatch case study (§7).
- **Extensions:** We outline stochastic and multi-stakeholder variants as initial extensions demonstrating the framework's flexibility (§6).

The remainder of the paper reviews related work (§2), develops the core and its guarantees (§3), the direct formulation (§4), the adaptive probes (§5) and the extensions (§6), reports the computational study (§7), discusses limitations (§8) and concludes (§9).

2. Background and related work

2.1. Pareto efficiency and post hoc *selection*

For a vector objective $F(x) = (f_1(x), \dots, f_m(x))$ (minimisation), x Pareto-dominates y if $f_i(x) \leq f_i(y)$ for all i with at least one strict inequality; the efficient set is the non-dominated set. Because it is typically large, a single recommendation is obtained by *post hoc* selection. The weighted sum, TOPSIS [12], compromise programming [32], knee selection [4], ELECTRE [24, 25], PROMETHEE [5] and hypervolume selection [33] are the standard tools; each requires weights, thresholds, a metric, or an already generated front. LexUR does not eliminate modelling choices. It relocates them into a declared family of admissible monotone probes and, on finite candidate sets, computes the resulting recommendation exactly while returning a labelled regret certificate.

2.2. *Achievement scalarising functions and ordered aggregation*

Achievement scalarising functions (ASF) project a reference point onto the efficient set through a parameterised metric interpolating between L_1 and L_∞ [30, 18]. Ordered weighted averaging (OWA) [31] and its extensions (WOWA) [28] connect ordered aggregation to lexicographic maximin.

Remark 1 (Recovery of standard scalarisations). The LexUR core recovers standard aggregation methods as exact special cases under specific, restricted probe families:

- **Weighted sum:** If $\mathcal{Q} = \{q_w(r) = w^\top r\}$, LexUR exactly minimises the weighted sum of normalised criteria.
- **Chebyshev / ASF:** If $\mathcal{Q} = \{q_w(r) = \max_i w_i r_i\}$, LexUR exactly minimises the Chebyshev distance to the ideal point.
- **Lexicographic minimax:** If $\mathcal{Q} = \{r_1, \ldots, r_m\}$ (the singleton family), LexUR performs a lexicographic minimax optimisation over the normalised regrets.

We do not claim LexUR subsumes these methods with its full adaptive family—there it is a *different* aggregation paradigm. Its distinguishing features are that it sorts *normalised regrets* (dimensionless disappointments relative to each probe's own attainable best and worst) over a *declared family of probes* and breaks ties through the whole sorted vector, and that it emits a certificate.

2.3. *Robust MCDA: SMAA, robust ordinal regression, preference robust optimisation*

The work closest to LexUR lies in *robust* multicriteria decision analysis and minimax regret optimisation [14]. **SMAA** [15, 27] computes, by Monte-Carlo integration over a distribution of admissible weights (and possibly criterion values), acceptability indices describing how strongly each alternative is supported. **Robust ordinal regression** [10, 9] reasons over the entire set of additive value functions compatible with a DM's holistic pairwise judgements, producing necessary and possible preference relations, sharing goals with inverse MCDA and the UTA family [13]. **Preference robust optimisation** [1, 11, 7, 29] optimises the worst-case expected utility over an ambiguity set of utility functions. LexUR shares the robust philosophy—perform well across a whole class of admissible preferences rather than commit to one—but differs in three ways. First, its admissible class is an explicit family of *monotone probes*, not a weight distribution or an elicited polyhedron. Second, its criterion is the *lexicographic minimax of normalised regret*, not an expected acceptability, a necessary/possible relation, or a single worst-case utility. Third, its output is an interpretable *certificate* naming the binding coalitions. We note that SMAA is inherently an *exploratory* framework designed to visualise the robustness of alternatives across weight spaces, rather than a single-choice prescriptive rule. By forcing it to recommend the alternative with the highest first-rank acceptability index (ties broken by expected value), we create a comparable baseline for §7; however, we emphasise that LexUR's single-shot default recommendation is meant to *complement* rather than compete with SMAA's rich exploratory indices. We likewise compare against a Savage minimax-regret (MMR) recommender over a sampled linear weight set (§7). More broadly, LexUR is complementary to data-driven preference disaggregation and inverse MCDA [8], which infer preference parameters from holistic judgements: LexUR instead dispenses with a fitted preference model and reasons over a declared class of questions, and could take an elicited probe family as input where such data exist.

2.4. Minimax regret and robust optimisation

Regret-based choice originates with Savage [26] and Loomes and Sugden [16]; minimax regret seeks alternatives minimising the largest shortfall from the best attainable value. Robust optimisation [2] and stochastic programming [3] address parameter uncertainty in the model rather than preference uncertainty in the DM. LexUR extends minimax regret in three directions: it is *lexicographic* (after the worst regret, minimise the second worst, and so on), *probe-based* (regret is computed over rational coalitions of criteria, not raw criteria), and *direct on a given candidate set* (computed without an additional scalarisation sweep, and front-free only in the structured continuous cases of §4).

2.5. Social choice and fairness

Aggregating criteria resembles social choice, but treating criteria as voters invites Condorcet cycles; LexUR avoids this by aggregating *regrets over declared questions*, not votes over criteria. For multi-stakeholder settings we draw on the maximin principle of Rawls [23] and the bargaining viewpoint of Nash [19]; we claim natural fairness properties (anonymity, monotonicity, worst-stakeholder protection), not a uniqueness theorem.

2.6. What is new relative to existing regret and robust-MCDA rules

While LexUR shares motivation with several robust approaches, its mathematical object and output differ fundamentally:

- **Versus minimax regret:** LexUR minimises a sorted vector of normalised probe regrets, not a single maximum regret over raw utilities or sampled weights.
- **Versus ASF/Chebyshev:** ASF commits to one scalarising function or one reference structure; LexUR evaluates a declared family and certifies binding probes.
- **Versus SMAA:** SMAA integrates over a preference distribution and reports acceptability; LexUR does not require a probability distribution and returns a leximax regret certificate.
- **Versus robust ordinal regression (ROR):** ROR reasons over compatible value functions after elicited comparisons; LexUR can operate without holistic preference judgements.
- **Versus preference robust optimisation (PRO):** PRO optimises worst-case utility over an ambiguity set; LexUR optimises lexicographically sorted normalised disappointments over interpretable monotone probes.
- **Versus ordered weighted aggregation (OWA):** LexUR sorts regrets across declared questions, not criterion values themselves. OWA specifies an ordered weight vector, whereas LexUR specifies a probe family; both therefore encode modelling choices in different forms. We include OWA utilities as a held-out evaluation family, not as evidence that LexUR subsumes OWA.

The novelty is not minimax regret alone, but lexicographic minimax over normalised disappointments induced by a declared monotone probe family, with an auditable certificate and an adaptive redundancy-aware probe construction.

Table 1 summarises where LexUR sits relative to representative MCDA and robust methods.

Table 1: Comparison of LexUR with representative selection and robust-MCDA methods.

Method	Required Input	Pareto Comp.	Output / Explanation	Key Limitation
Weighted sum	Weight vector	Yes ($w > 0$)	Single choice (Low)	Misses non-supported points in non-convex fronts
TOPSIS	Weights, distance metric	Not strictly	Single choice (Low)	Rank reversal; distance to non-feasible ideal
CP	Weights, L_p exponent	Yes ($p < \infty$)	Single choice (Low)	Non-linear scaling effects
ASF / Chebyshev	Reference point, scaling	Yes	Single choice (Low)	Requires an arbitrary reference point
MMR (sampled)	Linear weight distribution	Yes	Worst-case regret	Limited to linear aggregations
SMAA	Preference distribution	Usually	Acceptability indices	Exploratory; single-choice depends on post-processing
VIKOR	Weights, majority threshold	Not strictly	Single choice (Low)	Complex parameter interaction
Single-point HV	Reference point	Yes	Single choice (Low)	Box-volume score; sensitive to reference point
LexUR (ours)	Monotone probe family	Yes*	Leximax certificate	Sensitive to normalisation bounds

* Strict Pareto compatibility holds if probes separate criteria (Theorem 2).

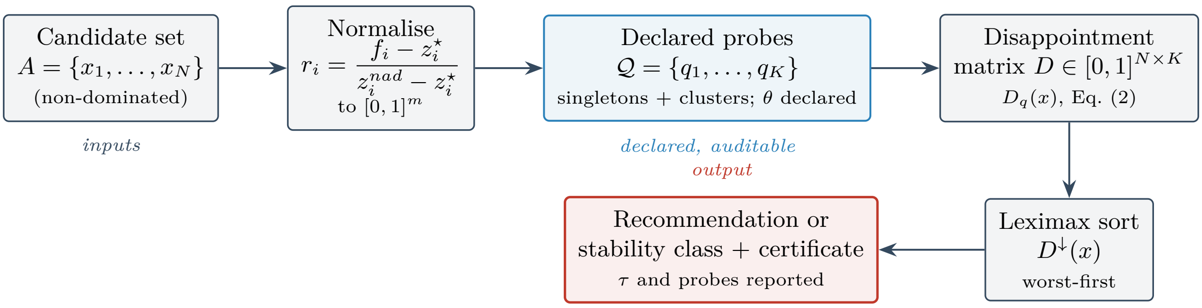

Figure 2: The LexUR pipeline. Declared inputs include the candidate set, normalisation bounds, probe family $\mathcal{Q}$, clustering rule θ, and numerical tolerance τ. Exact leximax comparison uses $\tau = 0$; numerical reporting returns a stability class unless a verified margin supports a unique recommendation. Every output retains the labelled regret certificate.

3. The Leximax Universal-Regret core

Figure 2 gives the end-to-end view of the rule developed in this section: from a candidate set, through normalisation and the declared probe family, to the leximax selection and its certificate.

3.1. Problem setting and normalisation

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be the feasible set and $F : \mathcal{X} \to \mathbb{R}^m$, $F(x) = (f_1(x), \dots, f_m(x))$, the vector objective.

Convention: All criteria f_i are assumed to be *minimised*. Maximisation objectives must have their sign reversed before normalisation. Consequently, lower normalised values r_i are better, and probes are non-decreasing functions of these costs.

For the finite case let $A = \{x_1, \dots, x_N\} \subseteq \mathcal{X}$ be a candidate set, e.g. an *a posteriori* approximation of the efficient set. Before evaluation, Pareto-dominated alternatives are removed from A unless the analyst explicitly requires selection over the raw un-filtered set. For the continuous case (§4) A is not required. Criteria are normalised to $[0, 1]$ by the positive affine map

$$r_i(x) = \frac{f_i(x) - z_i^\star}{z_i^{nad} - z_i^\star}, \qquad z_i^\star = \min_{x \in A} f_i(x), \quad z_i^{nad} = \max_{x \in A} f_i(x), \tag{1}$$

which attains both endpoints on A. A criterion is flagged as numerically degenerate if the range $z_i^{nad} - z_i^\star$ approaches zero. Degenerate criteria are reported and excluded from non-singleton probe construction unless the analyst supplies external bounds. If a criterion is degenerate over A, strict Pareto improvement on that criterion cannot occur inside A. If external bounds are supplied, singleton probes for degenerate criteria can remain. Excluding degenerate criteria does not weaken the strict Pareto compatibility for the remaining non-degenerate dimensions.

For the finite case all extrema ($z^\star$, z^{nad}, $q^\star$, q^{w} below) are taken over the active candidate set A; We use the active-set maximum over A, not the true nadir. It is a valid upper endpoint for the evaluated candidate set, but not necessarily a conservative bound for the underlying feasible problem unless A is known to contain the relevant nadir-attaining efficient points or external bounds are supplied. The rule is invariant to positive affine rescaling of the criteria when the bounds are transformed consistently; it is not invariant to arbitrary monotone transformations (e.g. $f_i \mapsto \log f_i$), which change the distances that define regret—so such transformations are a modelling choice the analyst must make deliberately. Because LexUR uses active-set normalisation, its recommendation is strongly set-dependent: adding a non-selected extreme candidate can change the normalisation and thus the recommendation. Consequently, LexUR is recommended only when the normalisation bounds are either fixed externally or reported through an interval/stability analysis.

3.2. Admissible probes

A *probe* is a scalar function $q : [0,1]^m \to \mathbb{R}$ encoding a rational decision question. The family $\mathcal{Q}$ is *declared*: it is part of the specification.

Definition 1 (Admissible probe)**.** *A probe q is* admissible *if (i) it is non-decreasing in each r_i (monotonicity) and (ii) $q(\mathbf{1}) > q(\mathbf{0})$, so it can be rescaled such that $q(\mathbf{0}) = 0$ and $q(\mathbf{1}) = 1$ (normalisation).*

Here $\mathbf{0}$ and $\mathbf{1}$ denote the all-zeros and all-ones vectors. The initial endpoint normalisation ($q(\mathbf{0}) = 0, q(\mathbf{1}) = 1$) serves to make values structurally scale-comparable across the heterogeneous family, before the empirical disappointment anchors ($q^\star, q^{\mathrm{w}}$) re-normalise the observed active range. Monotonicity gives *weak* Pareto compatibility; strict compatibility needs a separation property (Definition 2). Canonical admissible probes are the *singletons* $q_i(r) = r_i$; the *grand mean* $q_\mu(r) = \frac{1}{m}\sum_i r_i$ (the utilitarian question); the *grand maximum* $q_{\max}(r) = \max_i r_i$ (the egalitarian/worst-criterion question); and, for any coalition $S \subseteq \{1, \ldots, m\}$, the coalition mean and maximum. All have non-negative weights or are maxima, hence are monotone.

Definition 2 (Separating family)**.** *A probe family $\mathcal{Q}$* separates *criteria if for every i there is a probe $q \in \mathcal{Q}$ strictly increasing in r_i. Including all m singleton probes $q_i(r) = r_i$ is sufficient; LexUR always does so.*

Remark 2 (Probe multiplicity is a modelling choice)*.* The probe family is treated as a *deduplicated labelled set*: each declared question appears once. Duplicating a probe is *not* neutral. Because the leximax order sorts the full vector of probe-wise disappointments, a repeated probe contributes its disappointment more than once to the sorted vector and can therefore change lower-order lexicographic tie-breaking and the labelled certificate. Probe multiplicities thus act as implicit importance weights and must be interpreted as deliberate modelling choices, not as a harmless restatement of the same question. (The singleton probes are retained once each for separation, Definition 2.)

3.3. Disappointment and regret

For each probe $q \in \mathcal{Q}$ let $q^\star = \min_{x\in A} q(r(x))$ and $q^{\mathrm{w}} = \max_{x\in A} q(r(x))$ (or a user-supplied worst-acceptable bound, subject to $q^{\mathrm{w}} \geq \max_{x\in A} q(r(x))$ and $q^{\mathrm{w}} - q^\star \geq \varepsilon_q$). Define the active range $R_q =$

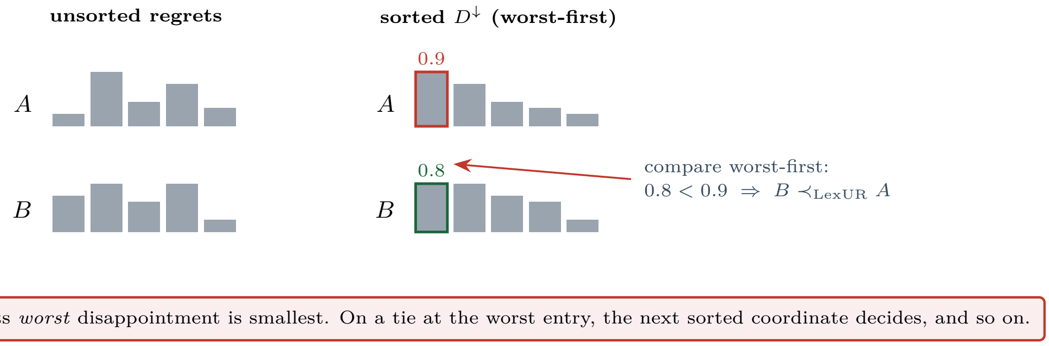

Figure 3: The leximax order. Each alternative's probe disappointments are sorted worst-first; alternatives are then compared lexicographically on the sorted vectors. Here B wins because its largest disappointment (0.8) is below A's (0.9); had they tied, the second-largest entries would decide, and so on.

$q^{\mathrm{w}} - q^{\star}$. If $R_q < \varepsilon_q$ for a small numerical threshold, the probe is degenerate and is excluded from the primary decision vector. Otherwise, the *normalised disappointment* of x under q is

$$D_q(x) = \frac{q(r(x)) - q^{\star}}{R_q} \in [0, 1], \tag{2}$$

so that $D_q(x) = 0$ iff x is q-optimal and 1 iff x attains the worst bound.

We use the term *disappointment* for the normalised shortfall of an alternative from the best attainable value of a single given probe, and *regret certificate* for the vector collecting these probe-wise disappointments. This is regret-like but differs from classical pairwise Savage regret unless the probe family is interpreted specifically as the uncertainty set of possible true preference functions.

3.4. The leximax order

Collect $D(x) = (D_{q_1}(x), \ldots, D_{q_K}(x))$, $K = |\mathcal{Q}|$, and sort it in *descending* order to obtain $D^{\downarrow}(x) = (D_{[1]}(x) \geq \cdots \geq D_{[K]}(x))$. The **Leximax Universal-Regret order**, a lexicographic minimax regret rule, is

$$x \preceq_{\text{LexUR}} y \iff D^{\downarrow}(x) \leq_{\text{lex}} D^{\downarrow}(y), \tag{3}$$

i.e. x is preferred if its worst disappointment is smaller; on ties, its second worst; and so on. We use the term *lexicographic minimax regret* (equivalently *leximax* of the sorted disappointments) consistently; this is exactly leximin applied to the negated disappointments, so the "leximax" and "leximin" descriptions coincide and we adopt the minimax-regret reading throughout. The *LexUR solution* is $x^{\text{LexUR}} \in \arg\min_x^{\text{lex}} D^{\downarrow}(x)$, and its *certificate* is the labelled, sorted vector $D^{\downarrow}(x^{\text{LexUR}})$ together with the probes attaining the largest entries. Figure 3 illustrates the comparison on a two-alternative example.

Exact LexUR defines a complete preorder, and we write $W_0 = \arg\min_{x \in A}^{\text{lex}} D^{\downarrow}(x)$ for its set of exact minimisers. In computation and practical reporting, a tolerance $\tau > 0$ is used to absorb estimation noise. For sorted vectors $a, b \in [0, 1]^K$, define $a \prec_\tau b$ if there exists $k = \min\{j : |a_j - b_j| > \tau\}$ and $a_k < b_k - \tau$. The τ-tolerance set is then

$$W_\tau = \{x \in A : \nexists y \in A \text{ such that } D^{\downarrow}(y) \prec_\tau D^{\downarrow}(x)\}.$$

We stress the precise status of these objects. W_τ is the set of τ-*undominated* alternatives under the relation $\prec_\tau$, which is irreflexive but *not* transitive; hence $\prec_\tau$ is neither a preorder nor does it induce an equivalence relation, and W_τ is *not* an equivalence class. We therefore avoid the terms "tolerance class" and "indifference class" for W_τ. When a uniquely stable recommendation is wanted, the appropriate object is the numerical stability set around an exact winner $x^{\star} \in W_0$, namely $C_\tau(x^{\star}) = \{x \in A :$

$D^{\downarrow}(x) \not\succ_\tau D^{\downarrow}(x^\star)\}$ (the alternatives not significantly worse than $x^\star$ under the declared tolerance), whose singleton-ness is guaranteed only under the margin condition of Theorem 3. The τ-tolerance rule is thus a numerical reporting and stability device around the exact or margin-separated winner, not a substitute preorder.

3.5. Interval bounds and the stability set

Because point estimates of normalisation bounds can cause frequent winner flips, true robustness requires evaluating the regrets over conservative interval bounds. We take $\mathcal{B}$ to be a genuinely *bounded* uncertainty box: each lower bound ranges in an interval $z_i^{\star,b} \in [\underline{z}_i^\star, \overline{z}_i^\star]$ and each upper bound in $z_i^{nad,b} \in [\underline{z}_i^{nad}, \overline{z}_i^{nad}]$, with the containment and range conditions

$$\overline{z}_i^\star \le \min_{x\in A} f_i(x), \qquad \underline{z}_i^{nad} \ge \max_{x\in A} f_i(x), \qquad z_i^{nad,b} - z_i^{\star,b} \ge \delta_i > 0 \quad \forall i,$$

so that

$$\mathcal{B} = \left\{ b = (z^{\star,b}, z^{nad,b}) : \ z_i^{\star,b} \in [\underline{z}_i^\star, \overline{z}_i^\star], \quad z_i^{nad,b} \in [\underline{z}_i^{nad}, \overline{z}_i^{nad}] \ \ \forall i \right\}$$

is a compact box. Boundedness guarantees that the suprema defining the worst-case envelope (Option C) are attained and finite. The containment conditions $\overline{z}_i^\star \le \min_A f_i$ and $\underline{z}_i^{nad} \ge \max_A f_i$ are required so that every normalised value

$$r_i^b(x) = \frac{f_i(x) - z_i^{\star,b}}{z_i^{nad,b} - z_i^{\star,b}}$$

remains in $[0,1]$ for all $x \in A$, which is the domain on which the probes $q : [0,1]^m \to \mathbb{R}$ are defined; the positive-range floor δ_i keeps the map well defined. (If one wishes to admit bounds that do not contain the active range, the domain of every q must be extended beyond $[0,1]^m$ explicitly.) For each bound scenario $b \in \mathcal{B}$ one then computes the disappointment $D_q^b(x)$, the sorted vector $D^{b,\downarrow}(x)$, and the τ-undominated set $W_\tau(b)$ as defined above.

The robust framework defines two fundamental stability sets:

1. **Necessary interval winner (Option A):** The intersection across all bounds:

$$S^\tau_{\text{necessary}} = \bigcap_{b\in\mathcal{B}} W_\tau(b).$$

2. **Possible interval winner (Option B):** The set of alternatives that win under some plausible bound:

$$S^\tau_{\text{possible}} = \bigcup_{b\in\mathcal{B}} W_\tau(b).$$

The necessary set is always a subset of the possible set. A unique robust recommendation exists if $W_\tau(b) = \{x^\star\}$ for all $b \in \mathcal{B}$. A uniform positive margin over $\mathcal{B}$ (Theorem 3 holding at every b) is a *sufficient* certificate of this property, but it is *not* necessary: the intersection $S^\tau_{\text{necessary}} = \bigcap_b W_\tau(b)$ can be a singleton even when some $W_\tau(b)$ contain extra alternatives and no uniform margin holds, and conversely a singleton necessary set does not by itself guarantee a unique recommendation across all bounds. When the necessary set is empty or the possible set contains several alternatives, the output should be reported as a stability set rather than a unique recommendation (and, per the remarks on W_τ above, not as an equivalence or indifference class).

For a single-shot operational robust default (Option C), one can compute the worst-case robust certificate:

$$\overline{D}_q(x) = \sup_{b\in\mathcal{B}} D_q^b(x),$$

and then select the recommendation by lexicographic minimisation of the sorted worst-case regrets $\overline{D}^{\downarrow}(x)$. Because the supremum is taken *separately* for each probe q, $\overline{D}(x)$ is a componentwise conservative envelope: its coordinates may be attained under different bound scenarios and need not be jointly realised under any single $b \in \mathcal{B}$. This worst-case robust default is a *separate*, conservative selection rule: its minimiser need *not* belong to the possible-winner set $S^{\tau}_{\text{possible}}$, and it should therefore be reported alongside the necessary and possible interval-winner sets rather than interpreted as an element of them. This is the familiar robust-optimisation phenomenon that a minimax compromise can be optimal under no individual scenario. For example, with a single probe and two bound scenarios b_1, b_2 giving disappointments (D^{b_1}, D^{b_2}) of $A{=}(0,1)$, $B{=}(1,0)$ and $C{=}(0.4, 0.4)$, we have $S^{\tau}_{\text{possible}} = \{A, B\}$, yet $\overline{D}(A) = \overline{D}(B) = 1 > \overline{D}(C) = 0.4$, so the robust default selects $C \notin S^{\tau}_{\text{possible}}$.

3.6. Foundational guarantees

Theorem 1 (Existence and completeness of exact LexUR)**.** *For any finite A and finite $\mathcal{Q}$, x^{LexUR} exists and exact $\preceq_{\text{LexUR}}$ is a complete preorder on A.*

Proof. For finite A and finite $\mathcal{Q}$, each $D^{\downarrow}(x) \in [0,1]^K$ is a well-defined real vector. The lexicographic order on $\mathbb{R}^K$ is reflexive, transitive and total, hence a complete preorder; since A is finite the lexicographic minimum exists. □

Theorem 2 (Weak and strict Pareto compatibility)**.** *Let x strictly Pareto-dominate y and let every $q \in \mathcal{Q}$ be monotone non-decreasing in each r_i. Then $D_q(x) \le D_q(y)$ for all q (*weak *compatibility), so $x \preceq_{\text{LexUR}} y$. If, in addition, $\mathcal{Q}$* separates *criteria (Definition 2)—in particular if $\mathcal{Q}$ contains the singleton probes—then $D_q(x) < D_q(y)$ for some q and $x \prec_{\text{LexUR}} y$.*

Proof. Let $x \prec_P y$, so $r_i(x) \le r_i(y)$ for all i with strict inequality for some j. For any monotone non-decreasing probe q, $q(r(x)) \le q(r(y))$; as $q^{\star}, q^{\text{w}}$ are constants independent of the alternative, $D_q(x) \le D_q(y)$ for every $q \in \mathcal{Q}$. We now invoke the monotonicity of order statistics: if $a_q \le b_q$ for every probe q, then the descending rearrangements satisfy $a^{\downarrow}_k \le b^{\downarrow}_k$ for all k (the k-th largest of a componentwise-smaller vector cannot exceed the k-th largest of the larger one). Applying this to $a_q = D_q(x)$ and $b_q = D_q(y)$ gives $D^{\downarrow}_k(x) \le D^{\downarrow}_k(y)$ for all k, hence $x \preceq_{\text{LexUR}} y$ (weak compatibility, monotonicity only). If $\mathcal{Q}$ separates criteria there is a probe q strictly increasing in r_j; provided this separating probe is non-degenerate over A (active range $q^{\text{w}} - q^{\star} \ge \varepsilon_q > 0$, which holds automatically whenever q strictly separates two alternatives of A), $q(r(x)) < q(r(y))$ gives $D_q(x) < D_q(y)$. The disappointment vector of x is then componentwise $\le$ that of y with at least one strict drop, so by the same order-statistic monotonicity (now with one strict inequality) the descending-sorted vectors satisfy $D^{\downarrow}(x) \le_{\text{lex}} D^{\downarrow}(y)$ strictly and $x \prec_{\text{LexUR}} y$. The counterexample in the main text shows the separation hypothesis cannot be dropped. □

The separation hypothesis is necessary: with only the probe $q(r) = r_2$, the dominator $r(x) = (0,1)$ and the dominated $r(y) = (0.1, 1)$ have equal disappointment, so monotonicity alone does not give a strict preference. LexUR always includes the singletons, so it separates.

Corollary 1 (Dominated exclusion)**.** *Let A be the active candidate set. If $x \in A$ is Pareto-dominated by some $y \in A$, and $\mathcal{Q}$ separates criteria, then x is not LexUR-optimal over A.*

Proof. If $x \in A$ is dominated by some $y \in A$ and $\mathcal{Q}$ separates criteria, Theorem 2 gives $y \prec_{\text{LexUR}} x$, so x is not a lexicographic minimiser over A. Without separation only $y \preceq_{\text{LexUR}} x$ holds and a dominated alternative may tie its dominator. □

Theorem 3 (Stability of the implemented winner under a margin condition)**.** *Let $\hat{D}_q(x)$ estimate $D_q(x)$ with $\eta = \max_{x,q} |\hat{D}_q(x) - D_q(x)|$. Suppose the exact winner x^{LexUR} beats every rival y at the first coordinate $k(y)$ at which their sorted vectors differ by more than τ, with margin $D_{[k(y)]}(y) - D_{[k(y)]}(x^{\text{LexUR}}) \ge$*

Δ*, and that all earlier coordinates agree within* $\tau - 2\eta$*. If* $2\eta < \min\{\Delta - \tau,\ \tau\}$*, the winner under* $\hat{D}$ *is unchanged.*

Proof. Write $\hat{D}_q(x) = D_q(x)+\xi_q(x)$, $|\xi_q(x)| \leq \eta$; sorting is 1-Lipschitz in $\|\cdot\|_\infty$, so each sorted coordinate moves by at most η. Fix a rival y and the deciding coordinate $k = k(y)$ with true margin $D_{[k]}(y) - D_{[k]}(x^{\mathrm{LexUR}}) \geq \Delta$. Under perturbation this margin is at least $\Delta - 2\eta$; since $2\eta < \Delta - \tau$ it stays above τ, so the τ-tolerance rule still decides coordinate k in favour of x^{LexUR}. For the earlier coordinates $j < k$, which agree within $\tau - 2\eta$ by hypothesis, the perturbed gaps stay within τ and are treated as ties, so the decision is not pre-empted before k. As $2\eta < \tau$ no spurious strict difference is created earlier. Hence x^{LexUR} still beats every rival and the winner is unchanged. (Exact comparison, $\tau = 0$, is excluded precisely because the second condition $2\eta < \tau$ then fails.) □

Theorem 3 is deliberately stated for the tolerance rule: exact lexicographic comparison is *not* perturbation-stable, because a winner that beats a rival only at a later coordinate after an exact earlier tie can be reversed by an arbitrarily small perturbation of the earlier coordinate. The tolerance τ absorbs exactly those near-ties. We note that the margin condition Δ in Theorem 3 can be restrictive on densely packed candidate sets, where the separation between certificates may be very small. Without that margin the theorem gives no winner stability guarantee; the appropriate output is therefore a tolerance or stability set rather than a uniquely stable recommendation. For continuous $\mathcal{X}$, existence requires compactness of $\mathcal{X}$ and continuity of each $q \circ r$; uniqueness is not guaranteed and a small set of tied exact minimisers may arise.

4. Exact candidate-set computation and conditional direct formulations

LexUR is primarily a candidate-set selection rule, and that is the setting of our experiments. We first present the exact finite algorithm. We then develop, more tentatively, how the recommendation can in principle be computed *directly* on a continuous feasible set, replacing the usual two-stage pipeline by a single robust search. We present the continuous case as a formulation with assumptions whose large-scale empirical validation we leave to future work.

4.1. Exact algorithm for finite candidate sets

The finite-set procedure is summarised in Figure 4.

The worst-case complexity is straightforward. With naive probe evaluation, winner selection costs $O(NKm + NK\log K + NK)$. Producing a complete lexicographic ranking of all candidates costs $O(NKm + NK\log K + NK\log N)$. Here $O(NKm)$ is the cost of probe evaluation and $O(NK\log K)$ is the cost of sorting the K-dimensional disappointment vector for each of the N candidates. The number of probes K is controlled by the construction of §5.

4.2. Cost of the anchors

Both the finite and continuous formulations need, for each probe, its attainable best $q^\star = \min_x q(r(x))$ and worst $q^{\mathrm{w}} = \max_x q(r(x))$. On a candidate set these are $O(N)$ scans; on a continuous $\mathcal{X}$ each is a scalar global optimisation, so the family contributes $2K$ such subproblems. Linear probes on a polyhedral feasible set yield linear anchor problems. For a general convex probe, minimising the probe may remain tractable, but maximising it over a convex set is not generally a convex optimisation problem. Non-convex feasible sets or probes may require global optimisation. Anchor computation can therefore be a substantial bottleneck and is not assumed cheaper than front approximation in all continuous cases.

Procedure (Figure 4): Exact finite-set LexUR ($\tau = 0$) and tolerance reporting

Input: Candidate set $A = \{x_1, \ldots, x_N\}$; objectives $F(x)$; labelled monotone probe family $\mathcal{Q} = \{q_1, \ldots, q_K\}$; degeneracy threshold ϵ_q.
Output: Exact minimiser set and labelled certificates; optionally a numerical τ-class.

1. Dominance filter: Remove any Pareto-dominated $x \in A$.
2. Normalisation: For each criterion i:

- Compute active range $z_i^\star = \min_A f_i(x)$ and $z_i^{nad} = \max_A f_i(x)$.
- If $z_i^{nad} - z_i^\star > \epsilon_q$, compute $r_i(x) = (f_i(x) - z_i^\star)/(z_i^{nad} - z_i^\star)$; otherwise flag the criterion and require external bounds or exclude probes that use it.

3. Anchor bounds: For each probe $q \in \mathcal{Q}$:

- Evaluate $V_{iq} = q(r(x_i))$ for all active $x_i \in A$.
- Find $q^\star = \min_i V_{iq}$ and $q^{\mathrm{w}} = \max_i V_{iq}$; flag and exclude a probe when $q^{\mathrm{w}} - q^\star \leq \epsilon_q$.

4. Regret evaluation: For each $x \in A$ and each valid probe $q \in \mathcal{Q}$:

- Compute $D_q(x) = (V_{iq} - q^\star)/(q^{\mathrm{w}} - q^\star)$.

5. Certificate sorting: For each x, sort the labelled pairs $(D_q(x), q)$ descending to obtain $D^{\downarrow}(x)$ without discarding probe names.
6. Exact selection: With $\tau = 0$, return every row attaining the lexicographic minimum.
7. Numerical reporting (optional): Around the exact minimiser $x^\star$, report W_τ or the stability set $C_\tau(x^\star)$ defined in §3; do not treat this tolerance relation as a preorder or call it an equivalence class.

Figure 4: Finite-set LexUR procedure.

4.3. Front-free formulation for structured cases

By formulating LexUR directly on $\mathcal{X}$, a front-free computation is theoretically available for structured cases (e.g., linear multiobjective problems), as we demonstrate empirically in §7. Lexicographic minimisation of the sorted regret vector over a continuous set $\mathcal{X}$ is exactly equivalent to sequentially minimising the cumulative sums of the p largest disappointments, $S_p(D(x)) = \sum_{k=1}^{p} D_{[k]}(x)$. For $p = 1, \ldots, K$, this admits an exact ordered-value linearisation [22]:

$$S_p(D(x)) = \min_{\alpha \in \mathbb{R}, u \in \mathbb{R}^K} \left[p\alpha + \sum_{q \in \mathcal{Q}} u_q \right] \quad \text{s.t.} \quad u_q \geq D_q(x) - \alpha, \quad u_q \geq 0 \quad \forall q \in \mathcal{Q}. \tag{4}$$

Lexicographic minimisation is implemented sequentially:

$$\gamma_1 = \min_{x \in \mathcal{X}} S_1(D(x)), \tag{5}$$

$$\gamma_p = \min_{x \in \mathcal{X}} S_p(D(x)) \quad \text{s.t.} \quad S_j(D(x)) \leq \gamma_j \quad \forall j < p. \tag{6}$$

If $\mathcal{X}$ is polyhedral and the probes are linear, each stage p can be solved as an LP (or MILP, if $\mathcal{X}$ contains integer variables) over the decision variables x, the scalar α, and the K auxiliary continuous variables u_q. Max-linear probes (e.g., $q(r) = \max_i w_i r_i$) are easily incorporated into this LP by enforcing $u_q \geq w_i r_i(x) - \alpha$ for all i.

However, because computing the per-probe anchor $q^{\mathrm{w}} = \max_{x \in \mathcal{X}} q(r(x))$ requires maximising a convex function over a convex set, these $2K$ global anchor subproblems may be no cheaper than front generation in the general case. In the continuous empirical demonstration (§7), there are $m = 3$ criteria and an adaptive family of $K = 7$ probes. Computing the $q^\star$ and q^{w} anchors requires $2K = 14$ global optimisation calls. The sequential lexicographic selection then evaluates up to $K = 5$ active solver calls (since degenerate probes are excluded), producing the reported 19 solver calls. We are careful not to overclaim: the core contribution of this paper remains the finite-set *post hoc* selection rule, and we do not claim LexUR as a general front-free mixed-integer solver.

5. Adaptive, redundancy-aware probe generation

5.1. The redundancy problem

The full coalition family (mean and maximum over every non-trivial subset of criteria) has size exponential in m. Worse, real criteria are often *redundant*: several measured objectives track the same underlying concern and are strongly positively correlated. Averaging selection rules implicitly *over-weight* such a group—it is counted once per member—biasing the recommendation.

Consider a 3-criteria numerical example where f_1 is a weak independent criterion and f_2, f_3 are redundant metrics (e.g. both measure environmental impact):

Alternative	r_1	r_2	r_3	Interpretation
A	0.70	0.20	0.21	weak independent criterion, good redundant group
B	0.45	0.35	0.36	balanced
C	0.25	0.55	0.56	good independent criterion, weak redundant group

A full coalition family overcounts $\{r_2, r_3\}$ (which are highly correlated), implicitly double-weighting the environmental impact. The adaptive clustered family reduces the number of redundant coalition probes, but singleton probes remain for separation. Therefore, it reduces but does not eliminate redundancy sensitivity. We need a probe family that (i) is small, (ii) stays monotone, and (iii) is less sensitive to redundancy than averaging aggregation.

5.2. Correlation clustering

We group *redundant* criteria by single-linkage connected components of the graph whose edges join pairs with Pearson correlation $\mathrm{Corr}(f_i, f_j) \geq \theta$. We explicitly state that both Pearson correlation and single-linkage clustering are deployed here as *empirical heuristics* rather than theoretically optimal redundancy measures. Correlations are computed over the un-normalised objective values $f_i(x)$ evaluated on the active candidate set A. Pearson correlation captures linear dependence but is known to be sensitive to outliers; rank-based alternatives (e.g., Spearman) or information-theoretic measures could be substituted in future work. Similarly, single-linkage is chosen because it treats redundancy as connected substitutability, though it is notorious for "chaining" effects that can over-merge criteria. To mitigate this, we report empirical sensitivity to θ and compare it with complete-linkage in the supplementary material.

Redundancy is positive correlation: two criteria that rise and fall together carry duplicate information, whereas negatively correlated criteria are genuine trade-offs and must remain separate—hence we threshold the signed correlation, not its absolute value. This yields clusters $C_1, \ldots, C_c$.

5.3. The adaptive probe family

LexUR always includes the canonical anchors—the m singletons, the grand mean, and the grand maximum—and adds one mean and one maximum probe *per cluster* of size > 1:

$$\mathcal{Q} = \{q_i\}_{i=1}^{m} \;\cup\; \{q_\mu, q_{\max}\} \;\cup \bigcup_{|C_k|>1} \{q_{C_k,\mathrm{mean}},\, q_{C_k,\max}\}. \tag{7}$$

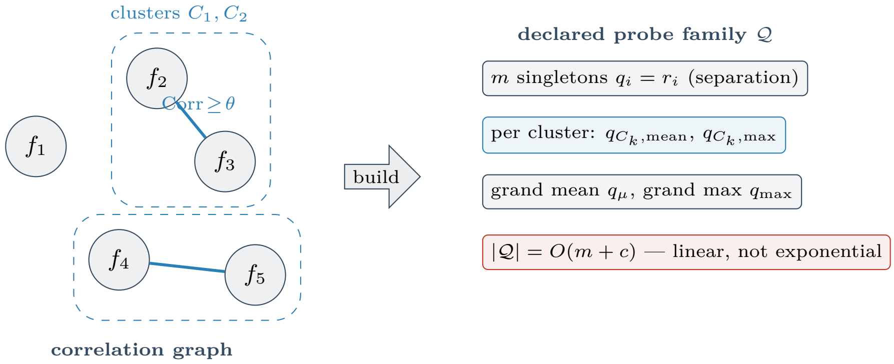

Figure 5: Adaptive probe construction. Positively correlated (redundant) criteria are grouped into clusters by thresholding the signed correlation at θ. Each nontrivial cluster contributes one mean and one max aggregate in place of the exponential subcoalition aggregates; singleton probes remain for separation. Together with the two grand probes, the family has $m + 2 + 2c$ probes, where c is the number of nontrivial clusters.

Figure 5 illustrates the construction from the correlation graph to the resulting declared family.

A redundant group thus contributes *one* aggregate (cluster) question rather than a separate aggregate per member, so the family does not amplify a correlated group the way an averaging aggregate does, while the count stays $O(m + c)$—linear, not exponential. We stress that this *reduces*, but does not eliminate, redundancy sensitivity: the singleton probes for the group's members remain in $\mathcal{Q}$ (they are needed for separation, Theorem 2), so a redundant group can still influence lower-order lexicographic tie-breaking. The claim is therefore empirical—less redundancy bias than averaging methods on the tested benchmarks (§7)—not a duplication-invariance theorem. All probes are non-negative-weighted sums or maxima, hence monotone, so Pareto compatibility (Theorem 2) is preserved.

Theorem 4 (Pareto compatibility for adaptive probes)**.** *If every probe is monotone, LexUR is weakly Pareto compatible. If, in addition, the family contains the singleton probes or otherwise separates criteria, LexUR is strictly Pareto compatible.*

Proof. If a probe q is a non-negative-weighted sum $q(r) = \sum_{i=1}^{m} w_i r_i$ with $w_i \geq 0$, then $r \leq r'$ componentwise implies $q(r) \leq q(r')$. Likewise, a maximum probe $q_{\max}(r) = \max_i w_i r_i$ satisfies $q_{\max}(r) \leq q_{\max}(r')$. Since all such probes are monotone non-decreasing, weak Pareto compatibility holds. The inclusion of singleton probes $q_i(r) = r_i$ ensures the family separates criteria, establishing strict Pareto compatibility as shown in Theorem 2. □

5.4. A perturbation bound (not a ranking guarantee)

We state the reduction effect as a regret-value perturbation lemma with explicit constants. We are deliberately modest: it bounds the change in individual disappointments, *not* the change in the LexUR ranking or winner, and it depends on the normalisation floor ε, which is decision-critical.

Lemma 1 (Disappointment perturbation)**.** *Suppose for a full-family probe q there is a reduced-family probe q' with $\sup_r |q(r) - q'(r)| \leq \eta$ (hence also $|q^\star - q'^\star| \leq \eta$ and $|q^{\mathrm{w}} - q'^{\mathrm{w}}| \leq \eta$), and let the normalisation ranges satisfy $q^{\mathrm{w}} - q^\star$, $q'^{\mathrm{w}} - q'^\star \geq \beta > 0$. Then for every x*

$$|D_q(x) - D_{q'}(x)| \;\leq\; \frac{4\eta}{\beta},$$

with the floor $\beta \geq \varepsilon$ in the worst case. The bound is vacuous when β is of order η, and informative only when the probe's attainable range is bounded away from zero—a regime the analyst can check directly.

Table 2: Probe-family size: full coalition family ($2(2^m - 1) - m$ probes) vs. adaptive family (concave instances, $N = 400$). The realised number of nontrivial clusters is $c = 0$ at every plotted value of m, so the adaptive count is $m + 2$; the redundant-objective experiment in Figure 11 evaluates actual clustering behaviour.

m	full coalitions	adaptive probes	reduction
3	11	5	2.2×
5	57	7	8.1×
8	502	10	50.2×
10	2,036	12	169.7×
15	65,519	17	3,854.1×

Proof. Fix q and its approximant q' with $\sup_r |q(r) - q'(r)| \leq \eta$; then also $|q^\star - q'^\star| \leq \eta$ and $|q^{\mathrm{w}} - q'^{\mathrm{w}}| \leq \eta$. Write the ranges $N = q^{\mathrm{w}} - q^\star$ and $N' = q'^{\mathrm{w}} - q'^\star$, both $\geq \beta > 0$. Using $D_q(x) \in [0, 1]$,

$$|D_q(x) - D_{q'}(x)| = \left| \frac{q(x) - q^\star}{N} - \frac{q'(x) - q'^\star}{N'} \right| \leq \frac{|q(x) - q'(x)| + |q^\star - q'^\star|}{N'} + D_q(x) \frac{|N - N'|}{N'} \leq \frac{2\eta}{\beta} + 1 \cdot \frac{2\eta}{\beta} = \frac{4\eta}{\beta}.$$

The constant is explicit and the dependence on the range floor β (worst case ε) is exhibited, making clear when the bound is informative. □

We emphasise that Lemma 1 provides a perturbation bound on the regrets, but it does *not* guarantee exact rank preservation of the alternatives. Thus, the probe reduction relies on theoretical bounds for safety but is ultimately an empirical heuristic whose efficacy is demonstrated in §7.

Whether the reduced family admits such an η-close q' for every full-family probe is *not* established by correlation clustering in general; we therefore present Lemma 1 as a sensitivity statement and rely on the empirical redundancy study (§7) for evidence that clustering retains decision quality.

Remark 3 (Residual redundancy sensitivity). Clustering does not establish duplication invariance. Singleton probes are retained for separation, so duplicating a criterion also duplicates coordinates in the sorted disappointment vector and can affect later lexicographic comparisons. Cluster-level probes reduce the empirical influence of correlated criteria, but the remaining sensitivity must be measured rather than assumed to vanish, even under perfect correlation.

Remark 4 (Correlation is not semantic redundancy, and clustering is set-dependent). Two further caveats bound the interpretation of the clustered family. First, positive correlation on A is *empirical redundancy*, not necessarily *semantic* redundancy: two criteria can be positively correlated because of the geometry of the feasible region rather than because they measure the same underlying concern. The clustering should therefore be read as empirical redundancy *detection*, not as a determination that two criteria encode the same preference. Second, because correlations are computed on the active candidate set A, the clusters—and hence the declared family $\mathcal{Q}$ itself—are set-dependent: adding or removing alternatives can change the correlation structure and thus change $\mathcal{Q}$. This is a second channel of set-dependence beyond the normalisation bounds discussed in §3, and it should be reported together with the normalisation stability analysis whenever the candidate set may change.

5.5. *Empirical reduction*

Table 2 (plotted in Figure 6) reports the realised reduction on concave instances. The full family here is $\{q_{\mathrm{mean},S}, q_{\mathrm{max},S} : \emptyset \neq S \subseteq \{1, \ldots, m\}\}$, of size $2(2^m - 1) - m$ (mean and max coincide on singletons); the adaptive family is linear in m, giving a nearly four-thousand-fold reduction by $m = 15$ with, as §7 shows, little loss of decision quality.

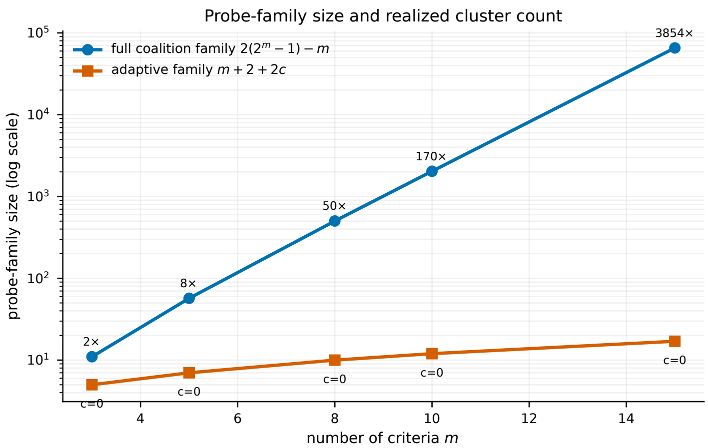

Figure 6: Probe-family size versus the number of criteria m (log scale). The full coalition family grows as $2(2^m - 1) - m$, whereas the adaptive correlation-clustering family has $m+2+2c$ probes. Here $c = 0$ throughout, so the plotted reduction comes from the restricted singleton-plus-grand-probe family rather than observed clustering. Clustering performance is evaluated separately in Figure 11.

5.6. Greedy probe selection (optional)

When even the adaptive family is larger than needed, a greedy set-cover heuristic selects a minimal subfamily distinguishing all dominant pairs; for the max-coverage variant it attains a $(1-1/e)$ guarantee and for full coverage a $1+\ln d$ guarantee [20, 6]. We do not use it in the experiments, which already run on the compact adaptive family.

6. Extension: stochastic regret certificates

When criteria are noisy, model $f_i(x)$ as drawn from a joint distribution $P(x)$. Each probe value is then a random variable; for non-linear probes (e.g. $q_{\max}$) its mean $\mu_q(x)$ is *not* a simple function of the criterion marginal means. Let ρ be a generic monotone risk functional over the scenarios, for example the sample average $\rho_q(x) = \frac{1}{S}\sum_{s=1}^{S} q(r(x, \omega^{(s)}))$ or the empirical Conditional Value-at-Risk $\rho_q(x) = \mathrm{CVaR}_\alpha(q(r(x, \omega)))$. The normalisation anchors must be taken consistently on this functional: $\rho_q^\star = \min_{x\in A} \rho_q(x)$ and $\rho_q^{\mathrm{w}} = \max_{x\in A} \rho_q(x)$. We define the *stochastic regret* as

$$D_q^\rho(x) = \frac{\rho_q(x) - \rho_q^\star}{\rho_q^{\mathrm{w}} - \rho_q^\star}, \tag{8}$$

assuming the active range is non-degenerate ($\geq \varepsilon_q$). Using CVaR_α penalises upper-tail outcomes of the probe value; it does not directly measure epistemic sample uncertainty unless the sampling procedure is explicitly modelled that way. Because CVaR is a monotone risk measure, exact Pareto compatibility is preserved.

To remove ambiguity about *what* is randomised and *where* the risk functional and anchors are applied, we fix the following conventions used throughout this section and in the experiments of §7. (i) The risk functional ρ is applied to the *probe value* $q(r(x, \omega))$—a scalar random variable per alternative and probe—not to the raw criteria and not to the already-normalised disappointments. (ii) The CVaR confidence level is $\alpha = 0.95$ in the reported experiments (the released config also sweeps $\alpha \in \{0.5, 0.8, 0.95\}$). (iii) The normalisation anchors are computed from the *same* risk functional on the *same* scenario sample as plug-in estimates, i.e. $\rho_q^\star = \min_{x\in A} \hat\rho_q(x)$ and $\rho_q^{\mathrm{w}} = \max_{x\in A} \hat\rho_q(x)$ with $\hat\rho_q$ the sample-average (or sample-CVaR) over the drawn scenarios; they are not taken from a separate population estimate or fixed externally unless the analyst supplies such bounds. This keeps the disappointment ratios well defined on the same sample on which the recommendation is computed.

Confidence-aware LexUR.. The variant evaluated in §7 (Figure 15) is a specific, reproducible instance of the above and we define it explicitly to avoid ambiguity. Given S i.i.d. observations per criterion with sample mean $\hat{\mu}_i(x)$ and sample standard deviation $\hat{\sigma}_i(x)$, and a one-sided confidence level γ with normal quantile z_γ ($z_{0.95} = 1.6449$ in our experiments), *confidence-aware LexUR* forms the upper-confidence objective (recall all criteria are minimised, so the conservative direction is upward)

$$\tilde{f}_i(x) = \hat{\mu}_i(x) + z_\gamma \frac{\hat{\sigma}_i(x)}{\sqrt{S}}, \qquad \gamma = 0.95,\ z_\gamma = 1.6449, \tag{9}$$

and then runs the *deterministic* finite LexUR rule on $\tilde{f}$. The normalisation anchors $z_i^\star, z_i^{nad}$ and the probe anchors $q^\star, q^{\mathrm{w}}$ are computed from these same plug-in upper-confidence objectives over the candidate set (i.e. option (iii) above), not from a separate population estimate; thus $r_i(x) = (\tilde{f}_i(x) - z_i^\star)/(z_i^{nad} - z_i^\star)$ with $z_i^\star = \min_{x \in A} \tilde{f}_i(x)$ and $z_i^{nad} = \max_{x \in A} \tilde{f}_i(x)$. With $z_\gamma = 0$ this reduces to plug-in *sample-average* LexUR; the term "confidence-aware" refers precisely to the statistical margin $z_\gamma \hat{\sigma}_i / \sqrt{S}$ in Eq. (9).

We flag one important limitation. Unlike sample-average LexUR and the monotone risk functionals (e.g. CVaR) of Proposition 1, the mean-plus-standard-error variant of Eq. (9) does *not* generally inherit scenario-wise Pareto compatibility, because the empirical standard-deviation term is not monotone under pointwise dominance. For instance, with one criterion and observations $x = (0, 100)$, $y = (101, 101)$, x dominates y in every scenario, yet x has far larger sample variance, so a sufficiently large z_γ gives $\tilde{f}(x) > \tilde{f}(y)$ and the rule may prefer the dominated y. We therefore present confidence-aware LexUR as an exploratory conservative heuristic; Proposition 1 (scenario-wise Pareto compatibility) is stated for sample-average and monotone-risk LexUR, and a practitioner who requires stochastic Pareto compatibility should use the CVaR (or another monotone risk) functional rather than the mean-plus-standard-error objective.

Proposition 1 (Scenario-wise Pareto compatibility)**.** *For finite A, finite $\mathcal{Q}$, and sample-average probe means $\hat{\mu}_q(x) = S^{-1} \sum_s q(r(x, \omega^{(s)}))$, if x weakly dominates y in every scenario and strictly dominates it in at least one criterion for at least one scenario, then the sample-average LexUR rule does not prefer y to x. If $\mathcal{Q}$ separates criteria and the dominance is strict on a separated coordinate, then x is strictly preferred by the exact LexUR order. We caution that this requires strict stochastic dominance across* all *scenarios, which is a very strong assumption as real stochastic problems often exhibit crossing dominance (where an alternative dominates in one scenario but is dominated in another).*

Proof. If x weakly dominates y under all scenarios and strictly dominates it in at least one criterion for at least one scenario, then $r(x, \omega^{(s)}) \leq r(y, \omega^{(s)})$ for all s, with strict inequality on at least one coordinate for some scenario. For any monotone probe q, $q(r(x, \omega^{(s)})) \leq q(r(y, \omega^{(s)}))$ for all scenarios. The functional $\rho_q(x) = \frac{1}{S} \sum_s q(r(x, \omega^{(s)}))$ preserves this weak inequality, meaning weak Pareto compatibility is preserved. If $\mathcal{Q}$ separates criteria and dominance is strict on a separated coordinate, then $\rho_q(x) < \rho_q(y)$ for that probe. Since the normalisation constants $\rho_q^\star$ and ρ_q^{w} are fixed across alternatives, the corresponding regret is strictly lower for x, giving $x \prec_{\mathrm{LexUR}} y$ by the logic of Theorem 2. □

Proposition 2 (Sample-average consistency under a margin condition)**.** *Let A and $\mathcal{Q}$ be finite, and assume non-degenerate population probe ranges with a strictly positive range floor, $\rho_q^{\mathrm{w}} - \rho_q^\star \geq \beta > 0$ for every $q \in \mathcal{Q}$ (so the disappointment ratios are bounded and stable). For each $x \in A$ and $q \in \mathcal{Q}$, suppose $q(r(x, \omega))$ is i.i.d. across scenarios with finite variance. Let $\mu_q(x) = \mathbb{E}[q(r(x, \omega))]$, and let $\hat{\mu}_q(x)$ be its sample average. Assume uniform convergence of the sample probe values to their population means. Define population and sample disappointment vectors using a clearly defined stochastic normalisation (e.g., sample-average anchors). If the population tolerance-LexUR winner satisfies the positive margin condition of Theorem 3, then the sample-average LexUR winner converges in probability to the population* *winner as $S \to \infty$.*

Proof. Since A and $\mathcal{Q}$ are finite, the weak law of large numbers gives uniform convergence in probability of $\hat{\mu}_q(x)$ to $\mu_q(x)$. Because the minima and maxima over finite sets are continuous operations, the estimated anchors $\hat{q}^\star$ and $\hat{q}^{\mathrm{w}}$ converge in probability to the population anchors. Since the population range satisfies $\rho_q^{\mathrm{w}} - \rho_q^\star \geq \beta > 0$, the denominator is bounded away from zero, so the disappointment ratio is a continuous function of the (converging) numerator and denominator; therefore the sample disappointment vectors converge uniformly in probability to the population disappointment vectors. Under the margin condition of Theorem 3, there exists an $\eta > 0$ such that perturbations strictly within η do not change the tolerance-LexUR winner. Since the probability of the maximum estimation error exceeding η vanishes as $S \to \infty$, the sample-average LexUR winner converges in probability to the population winner. $\square$

§7 gives an empirical confirmation.

7. Computational study

7.1. Design and protocol

Software Architecture and Reproducibility.. All experiments are implemented in the accompanying modular Python reproducibility package. Core selection methods (`lexur.methods`), performance metrics (`lexur.metrics`), and test problems (`lexur.problems`) are separated. A recommendation and its labelled disappointment certificate are obtained directly from an objective matrix `F`:

```
from lexur.methods import lexur
winner, D, labels, _ = lexur(F, return_detail=True)
certificate = sorted(zip(labels, D[winner]),
                     key=lambda item: item[1], reverse=True)
```

The protocol uses `pyarrow` for Parquet persistence and records the configuration, scientific-source, and environment identity of each run. A single command executes the frozen pipeline; failed and missing gates remain in the generated report.

Section 7.2 reports the main robust benchmark (7,200 instances) using this frozen broadened audit protocol. Sections 7.3–7.9 report the mechanistic analysis on a controlled 480-instance study used to understand behaviour, redundancy, and sensitivity.

We evaluate selection *quality* under a held-out protocol: each method picks one alternative using only its own machinery, and the pick is scored against preference models that *no* method used during selection. To avoid biasing the evaluation toward the additive/scalarising logic of the probes, the held-out set spans **six** families—random linear $-w^\top r$, weighted Chebyshev $-\max_i w_i r_i$, augmented ASF, CES, a *non-additive* Choquet-inspired 2-additive max-interaction utility with random non-negative masses on singletons and pairs (Appendix A), and a *satisficing* threshold utility penalising exceedances of random aspiration levels. Weights are drawn from a uniform simplex; 300 utilities are sampled per family per replication. We report two non-negative, lower-is-better losses, each normalised per utility against the best and worst feasible alternatives. Formally, for a held-out utility u (higher is better) with in-sample best $x_u^\star \in \arg\max_{x\in A} u(x)$ and worst $x_u^- \in \arg\min_{x\in A} u(x)$, the *held-out loss* of a recommended alternative x is

$$L_u(x) = \frac{u(x_u^\star) - u(x)}{u(x_u^\star) - u(x_u^-)} \in [0, 1], \tag{10}$$

so $L_u(x) = 0$ when x is u-optimal on A and 1 when it is u-worst (a small ε in the denominator guards degenerate utilities). The *mean held-out loss* averages $L_u(x)$ over all sampled utilities and families; the *tail held-out loss* is $\mathrm{CVaR}_{10\%}$, the average of $L_u(x)$ over the worst 10% of utility draws, and serves as our robust tail measure. We intentionally report both mean loss and tail loss, emphasizing that robust methods will typically underperform on average in exchange for bounding the worst-case disappointment.

Candidate sets of $N = 300$ non-dominated points are generated for four front geometries (linear, concave, convex, disconnected, following the standard DTLZ families) with $m \in \{3, 5, 8, 10\}$, over 30 independent replications (480 instances). Points are sampled uniformly from the theoretical fronts, and any mutually dominated points are removed. The exact same candidate sets and held-out utility draws are shared across all methods within each replication to ensure paired comparisons. **Baselines and parameters.** TOPSIS, compromise programming (CP, p=2), knee selection (distance-to-hyperplane), achievement scalarisation (ASF, reference at the ideal), and a random-weight popularity rule (RW) all use equal weights where applicable. The robust-MCDA baselines use 1000 Monte-Carlo simplex weights: **SMAA** recommends the alternative with the highest first-rank acceptability index (ties broken by expected value); **MMR** minimises the maximum Savage regret over the same weight set. LexUR uses the adaptive probe family with $\theta = 0.6$ and the tolerance-based leximax rule. Significance is assessed by the Friedman test with Nemenyi critical difference (CD) and by Holm-corrected Wilcoxon signed-rank tests against LexUR with Cliff's δ effect size.

7.2. Main Benchmark: The Broadened Protocol (7,200 instances)

To guard against confirmation bias we additionally ran a frozen audit protocol (frozen config, single-command reproduction) on a deliberately broadened suite: **11** methods (adding VIKOR, single-point hypervolume, and Chebyshev minimax regret), **ten** held-out families spanning additive *and* non-additive preferences (linear, Chebyshev, augmented ASF, CES, Choquet, satisficing, sparse-linear, L_p, OWA, and max-min), and eight geometries (the four canonical fronts—linear, convex, concave, disconnected—plus degenerate, asymmetric, many-knee, and irregular variants). We define one *paired instance* as a tuple (candidate-set size N, m, geometry, replication): every method selects on the identical candidate set and is scored against the identical held-out utility draws within each instance. The protocol sweeps $N \in \{100, 300, 1000\}$, $m \in \{3, 5, 8, 10, 15, 20\}$, the eight geometries, and 50 replications, giving

$$\underbrace{3}_{N} \times \underbrace{6}_{m} \times \underbrace{8}_{\text{geom.}} \times \underbrace{50}_{\text{rep.}} = 7,200 \text{ paired instances.}$$

A Dirichlet weight shape $\alpha \in \{0.2, 1, 5\}$ (sparse, uniform, balanced) is assigned to each replication by deterministic cycling ($\alpha = \alpha_{\text{rep mod 3}}$, so the three shapes are nearly balanced—$17, 17, 16$ across the 50 replications—within each $(N, m, \text{geometry})$ cell) rather than forming a separate multiplicative factor, and within each instance the per-instance loss is averaged over all ten held-out families and their sampled utilities. With 11 methods and $n = 7,200$ paired observations, the Nemenyi critical difference is $\text{CD} = 0.178$. The protocol fixes, in advance, a set of *acceptance gates*, each tied to one precise claim; we report the outcome verbatim in Table 3 rather than only favourable numbers.

The empirical benchmark evaluates *point-bound LexUR*; interval-LexUR is formalised as a robustness reporting layer (§3) but not benchmarked here as a separate recommender. For baselines, VIKOR uses standard $v = 0.5$, ChebMMR evaluates minimax regret over the Chebyshev metric, and SMAA recommends the highest first-rank acceptability index. Because this comparator obtains the best average rank on this suite we specify it precisely. To avoid a common terminological pitfall, we do *not* call it "hypervolume contribution": the rule selects the alternative maximising the *single-point* hypervolume of the singleton $\{x\}$ with reference point $\mathbf{1}$, i.e. the box volume

$$\text{HV}_1(x) = \prod_{i=1}^{m} \max\big(1 - r_i(x), 0\big),$$

Table 3: Frozen audit acceptance gates (reproduced at full scale): 9 PASS, 1 CHECK. Quality degradation refers to the mean tail-loss increase under perturbation. The CHECK result highlights a non-negligible degradation (0.153, more than three times the pre-registered 0.05 threshold) together with frequent winner-identity flips under nadir error.

Claim	Gate	Result	Evidence
Pareto compatibility	Dominated selections = 0	PASS	0/150 injections
Scale invariance	Positive-affine invariance	PASS	100% identical
Stability under bounds	Nadir-error stability	CHECK	degr. 0.1533 (thr. 0.05)
Competitive w/ ASF	Non-inferior vs ASF (tail)	PASS	CI upper 0.0015 < 2% margin
Competitive w/ MMR	Non-inferior vs MMR (tail)	PASS	CI upper 0.0021 < 0.01 margin
Beats classical methods	Beats baselines (tail)	PASS	δ: TOPSIS .17, CP .07, SMAA .61
Redundancy correction	Redundancy < averaging	PASS	0.256 vs 0.359
Adaptive sufficiency	Adaptive $\approx$ full (tail gap ≤ 0.01)	PASS	gap 0.0053
Continuous applicability	Direct computation (LP + MILP)	PASS	19 vs 150; 4 vs 60 solver calls
Stochastic safety	Stochastic LexUR tail loss	PASS	risk 0.383 vs 0.383

not the exclusive hypervolume contribution $\mathrm{HV}(A)-\mathrm{HV}(A\setminus\{x\})$, which differs because dominated boxes overlap. We denote this baseline HV throughout and read it as the single-point hypervolume score. It operates on the *same* $[0,1]$ active-set normalisation as every other method (Eq. 1); the reference point is the normalised nadir $\mathbf{1}$; ties are broken by the first index attaining the maximum; and no separate extreme-point protection is applied (the box-to-reference score already rewards near-ideal coordinates). Like LexUR, HV therefore inherits sensitivity to the active-set normalisation and to the reference point.

On this harder suite LexUR has tail-loss rank 4.67. Within the Nemenyi critical difference (CD = 0.178), LexUR is indistinguishable from MMR (4.57), VIKOR (4.71), and ASF (4.77), while HV has a better average rank (4.34) beyond one CD. The remaining ranks are ChebMMR 5.03, CP 5.28, TOPSIS 6.83, knee 7.32, RW 9.21, and SMAA 9.26 (Figure 7). That single-point HV achieves the best average rank does not displace LexUR's contribution, and we report it openly. HV returns a bare point with no semantic explanation, whereas LexUR emits a labelled certificate of the binding probes and explicit stability diagnostics; HV depends strongly on the reference point and its product score rewards multiplicative balance across criteria, a different and implicit value judgement; and LexUR remains within one CD of the strongest *robust* scalarisations (MMR, VIKOR, ASF) on the tail metric. The value of LexUR is therefore not that it tops the average-rank table but that it is a robust finite-candidate selector with an auditable, labelled regret certificate. A frozen *non-inferiority* test makes the comparison precise. For each baseline we define the paired difference $\Delta = \mathrm{loss}_{\mathrm{LexUR}} - \mathrm{loss}_{\mathrm{baseline}}$ on tail held-out loss (so $\Delta > 0$ means LexUR is worse), and declare LexUR non-inferior if the upper end of the 95% bootstrap confidence interval of Δ falls below the pre-registered margin. The margin is fixed in advance in the frozen config as max(0.01 absolute, 2% relative to the baseline loss). Confidence intervals come from a paired bootstrap with 10,000 resamples drawn at the level of the paired instance $(N, m, \mathrm{geometry}, \mathrm{replication})$ over all $n = 7,200$ instances, and the two non-inferiority tests (versus ASF and versus MMR) are Holm-corrected. The upper 95% CI bounds are 0.0015 (ASF) and 0.0021 (MMR), both below margin, with negligible effect size (Cliff's $\delta \leq 0.015$)—supporting "practically equivalent to the best robust scalarisations" rather than "better". Direct computation is demonstrated on a continuous linear MOP, where LexUR via sequential LPs uses 19 solver calls and *no* front enumeration vs. 150 for generate-then-select, at equal or better tail quality. On a secondary MILP test, direct computation resolves the exact answer in just 4 solver calls compared to 60 for exhaustive generation. The confidence-aware stochastic LexUR attains near-zero out-of-sample worst-criterion risk (≈ 0) against 0.28 for TOPSIS and 0.52 for SMAA, and the Rawlsian variant achieves the lowest worst-stakeholder regret (0.362) and inequality (Gini 0.207) among average-, Nash- and max-min rules.

We state the single CHECK outcome on Gate 3 (Nadir-error stability) plainly: the pre-specified nadir-stability gate was *not met*. The observed degradation of 0.1533 is more than three times the pre-registered

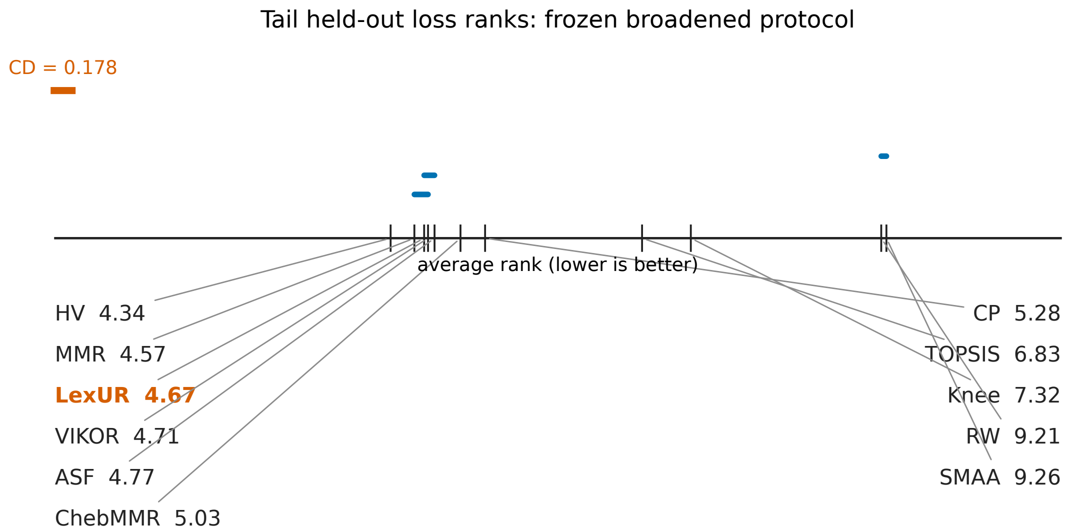

Figure 7: Average ranks on tail held-out loss over the broadened protocol suite (11 methods, 10 families, 8 geometries, $7,200$ paired instances; Nemenyi CD = 0.178). LexUR lies within one CD of MMR, VIKOR, and ASF, but not HV; its paired non-inferiority results versus ASF and MMR are reported separately.

threshold of 0.05, so this is not a marginal miss. We nonetheless report it openly because, although the gate fails on the strict identity-stability criterion, the resulting loss degradation remains acceptable in many cases (Table 7); what is genuinely sensitive is the *identity* of the recommendation rather than its held-out quality. The gate flags exactly this: LexUR's recommendation identity can be sensitive to incorrect point estimates of the nadir, differing from Table 7 mainly in that the protocol's strict hard-coded stability threshold (0.05 mean degradation) was tripped by the 0.1533 empirical degradation. For practitioners, this CHECK explicitly means that if the true nadir is unknown or highly uncertain, they should not rely on naive point estimates, but must instead supply conservative interval bounds (evaluating worst-case regret over a box of plausible nadirs) to ensure true robustness, as further discussed in §8.

7.3. Mechanistic Analysis: The Controlled 480-instance Study

Table 4 reports mean and tail loss averaged over all geometries and m. On *tail* loss the four robust/compromise methods MMR (0.470), ASF (0.472), LexUR (0.481) and CP (0.504) are close, and all clearly beat the distance-based and popularity/Monte-Carlo single-choice rules: LexUR reduces tail loss by 10.8% relative to TOPSIS (0.539), 20.6% relative to knee (0.606), 46.6% relative to RW (0.900) and 47.4% relative to the SMAA-derived single-choice rule (0.914). Figure 9 shows the average-rank CD diagram (Friedman $p < 10^{-15}$, $n = 480$, CD = 0.48). Read rigorously: **LexUR groups with CP** (rank gap 0.02). LexUR is close to MMR and ASF with negligible paired effect sizes (Cliff's $|\delta| \leq 0.034$), but just outside the Nemenyi CD in the controlled study (gaps 0.59 and 0.50 respectively). Against the methods practitioners actually use, LexUR is significantly better with small-to-large effect: $\delta = 0.087$ (TOPSIS), 0.285 (knee), 0.850 (RW), 0.873 (SMAA), all $p < 10^{-4}$. LexUR is clearly ahead of the default distance-based methods. The SMAA-derived first-rank recommender performs poorly on our tail-loss metric, but this should not be interpreted as a general defeat of SMAA, whose purpose is exploratory robustness analysis rather than single-point prescription. We therefore claim only that LexUR clearly outperforms the classical distance-based and popularity-style single-choice baselines under the tail-loss metric, while remaining competitive with ASF, MMR and CP—not that it dominates them. (Note that MMR is itself a sampled Monte-Carlo regret rule and CP a compromise/averaging-style rule, both close to LexUR; the gap is to the *distance-based and popularity* rules, not to robust sampling *per se*.) On *mean*

Table 4: Held-out loss over four geometries × $m \in \{3, 5, 8, 10\}$, 30 replications (480 instances; lower is better). Tail loss is the worst-case ($\mathrm{CVaR}_{10\%}$) metric. Best in each column in **bold**; on tail loss LexUR is within a negligible effect size ($|\delta| \le 0.034$) of MMR/ASF/CP.

Method	mean loss	tail loss
TOPSIS	0.220	0.539
CP	**0.207**	0.504
Knee	0.308	0.606
RW	0.462	0.900
ASF	0.234	0.472
SMAA	0.468	0.914
MMR	0.228	**0.470**
LexUR	0.245	0.481

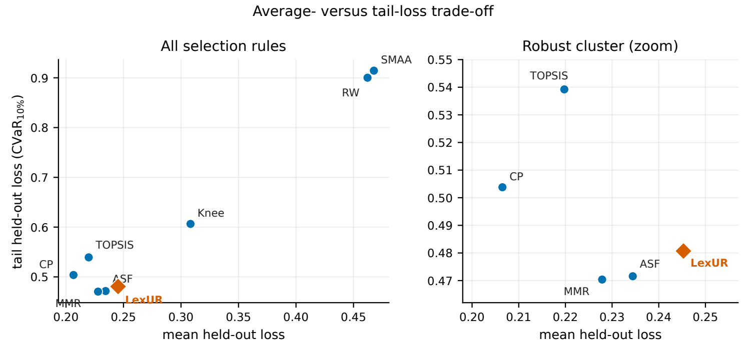

Figure 8: Average- versus worst-case held-out loss across selection rules (480-instance mechanistic study; lower-left is better). LexUR sits in the robust cluster with ASF, MMR and CP on worst-case (tail) loss while the random-weight popularity and SMAA-derived single-choice rules (RW, SMAA) trade away worst-case protection. The right panel enlarges the robust cluster.

loss the averaging methods (CP, MMR) lead and LexUR is mid-pack—the expected price of worst-case robustness (§8). Figure 8 makes this average-versus-worst-case structure explicit: LexUR lies in the robust lower-left cluster, well separated from the averaging/Monte-Carlo rules.

7.4. Behaviour across preference families

Table 5 (visualised as a heatmap in Figure 10) breaks the loss down by held-out family on a representative setting (concave, $m = 8$), including the two non-additive families. The structure is a clear trade-off: averaging methods (TOPSIS, SMAA) are strong on linear utilities but weak on the max-type families; ASF and MMR are the reverse—best on Chebyshev/CES but worst on linear. LexUR sits between the two camps: it removes much of ASF's linear-utility weakness (0.598 vs 0.674) while staying competitive on the max-type *and* the non-additive Choquet (0.368) and satisficing (0.468) families. We do *not* claim LexUR is the flattest profile—on this setting CP is flatter and has the lowest overall mean (0.404)—only that LexUR is robust across additive and non-additive preferences without being tuned to any.

7.5. Certificate regret and uniformity

By construction LexUR minimises the leximax of its certificate regret. Empirically (concave, $m = 8$) LexUR attains worst-case certificate regret 0.708, far below every alternative (MMR 0.921, CP 0.936, ASF 0.953, TOPSIS 0.974), with one of the most uniform regret profiles (sorted-disappointment std 0.219, second to MMR's 0.207). This is by design—LexUR optimises exactly this object—so it is a consistency check, not an independent win; its decision value is that the recommendation is *auditable*.

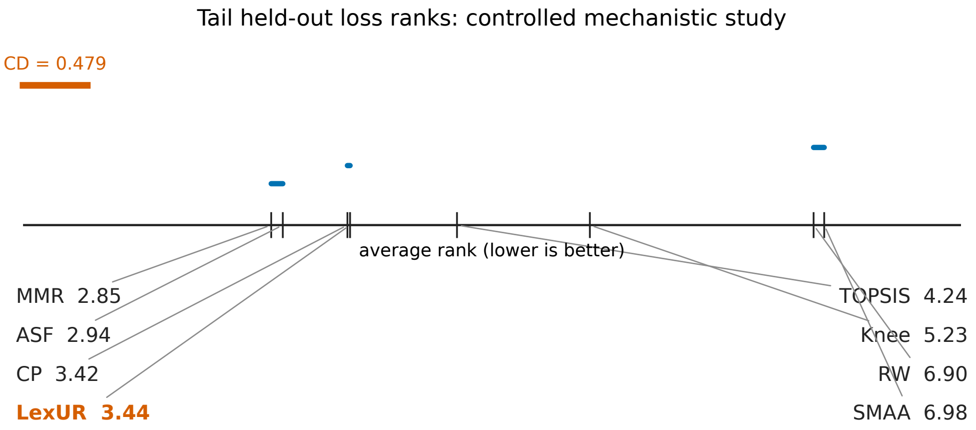

Figure 9: Average ranks on tail (worst-case) held-out loss with Nemenyi critical difference ($n = 480$, $\alpha = 0.05$, CD $= 0.48$). LexUR groups with CP. MMR and ASF have slightly better average ranks, with gaps 0.59 and 0.50 that lie just beyond the CD, although their paired effect sizes relative to LexUR are negligible. LexUR outperforms the SMAA-derived first-rank single-choice rule used here.

Table 5: Held-out loss by preference family (concave, $m = 8$, 30 replications), including the non-additive Choquet and satisficing families. Lowest per column in **bold**. The table illustrates the averaging-vs-worst-case trade-off; LexUR is balanced across additive and non-additive families. The *augmented ASF* family is the weighted Chebyshev term plus a small (0.05) linear augmentation on the same weights, so it induces almost the same induced ranking—and hence almost the same held-out loss—as the Chebyshev column; the two coincide to three decimals except where the augmentation flips a ranking (e.g. MMR, 0.442 vs 0.443).

Method	linear	Cheby.	aug. ASF	CES	Choquet	satisfice	overall
TOPSIS	**0.391**	0.511	0.511	0.563	0.341	0.511	0.471
CP	0.486	0.458	0.458	0.440	0.307	**0.278**	**0.404**
Knee	0.477	0.503	0.503	0.524	0.479	0.570	0.509
RW	0.403	0.519	0.519	0.569	0.379	0.552	0.490
ASF	0.674	0.434	0.434	**0.329**	**0.288**	0.388	0.425
SMAA	0.398	0.521	0.521	0.575	0.367	0.546	0.488
MMR	0.653	0.442	0.443	0.348	0.344	0.402	0.439
LexUR	0.598	0.454	0.454	0.374	0.368	0.468	0.453

7.6. Robustness to redundant criteria

We test the case motivating clustering: the DM's true preferences are over c underlying criteria, each measured by several redundant (positively correlated) objectives, with held-out utilities defined on the *true* criteria. The clustering recovers the redundancy groups essentially exactly. Table 6 and Figure 11 (four group structures $\times$ three geometries, 30 replications) show close grouped-loss point estimates for LexUR and ASF and place LexUR among the strongest methods on tail loss; the averaging and popularity-style methods (TOPSIS, RW, SMAA), which amplify redundant groups, are 19–35% worse on grouped loss (TOPSIS 0.332, SMAA 0.413 vs LexUR 0.268). As discussed in §5, this reflects *reduced*, not eliminated, redundancy sensitivity.

7.7. Distinctness, dominated exclusion, ablation, sensitivity

LexUR and the SMAA recommender coincide on 1.9% of instances, with the same mean tolerance-set Jaccard overlap, and select alternatives at a mean RMS normalised criterion distance of 0.351: LexUR is operationally a different rule, not a re-description of acceptability analysis. Across 200 randomised trials with injected dominated points (for which the standard Pareto-dominance filter was explicitly

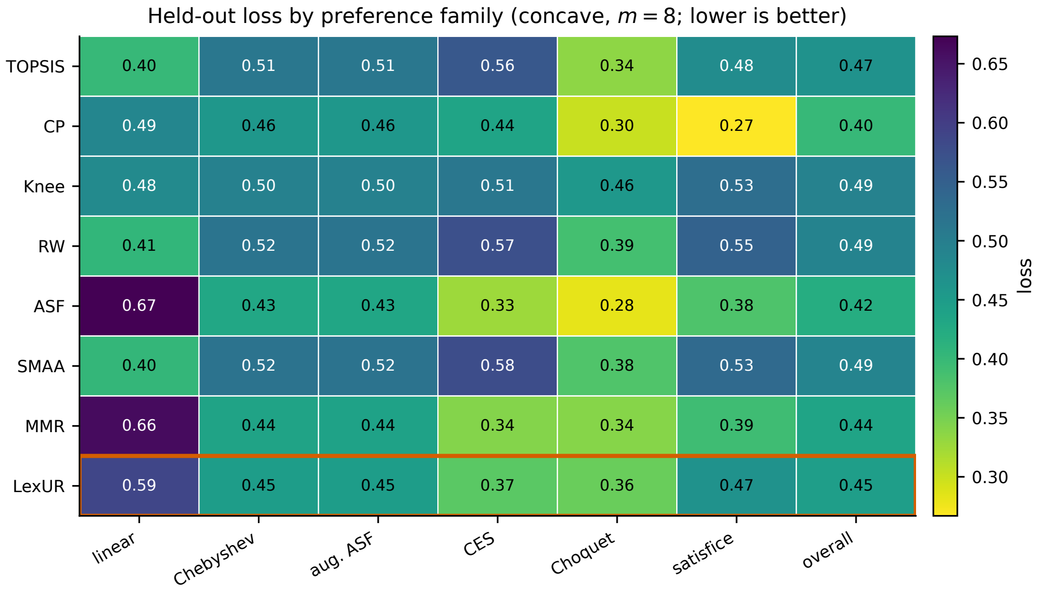

Figure 10: Held-out loss by preference family (concave, $m = 8$); lower is better and colour encodes magnitude using a perceptually uniform scale. Averaging rules (TOPSIS, SMAA) are strong on linear utilities but weak on max-type families; ASF and MMR are the reverse. LexUR (outlined) occupies an intermediate profile across additive and non-additive families; CP has the lowest overall mean, and no undefined "most balanced" claim is made.

Table 6: Redundant-objective benchmark (true preferences over the underlying criteria; 4 group structures × 3 geometries, 30 replications). Grouped loss measures de-redundification quality. Lower is better; best in **bold**.

Method	grouped loss	tail loss
TOPSIS	0.332	0.455
CP	0.317	0.423
Knee	0.322	0.489
RW	0.409	0.598
ASF	**0.266**	0.406
SMAA	0.413	0.603
MMR	0.295	**0.387**
LexUR	0.268	0.391

disabled), LexUR never selected a dominated alternative (0/200), confirming Corollary 1. Ablation (concave, $m = 8$): dropping the singletons raises loss ($0.456 \to 0.483$) and worst-case certificate regret ($0.701 \to 0.783$); a max-only family lowers mean loss to 0.420 but inflates certificate regret to 0.929, defeating the robustness aim; clustering is, by design, a no-op when criteria are independent (its value is the redundancy result); Figure 12 summarises the ablation, and Figure 13 the LexUR–SMAA distinctness across geometries and m.

LexUR is insensitive to the clustering threshold θ over $[0.3, 0.9]$ across all four geometries. Nadir perturbations reveal an important practical limitation. Although many changed winners remain close in held-out loss, the identity of the LexUR recommendation can change frequently because the leximax certificate often contains near-ties. In the frozen audit protocol, the nadir-error stability gate was marked CHECK rather than PASS. Reliable normalisation bounds therefore remain a practical condition for using LexUR, and users should report both winner-change rates and loss degradation under plausible bound perturbations. Table 7 and Figure 14 quantify these diagnostics.

7.8. Runtime scalability

LexUR achieves its robustness without Monte-Carlo preference sampling. On our evaluation hardware, evaluating a candidate set instance ($N = 300$, $m = 8$) takes under 10 milliseconds for LexUR,

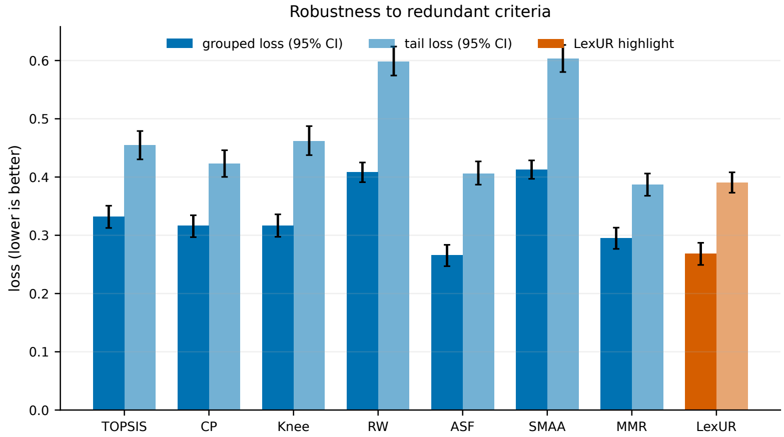

Figure 11: Robustness to redundant criteria. On grouped loss (true preferences over the underlying criteria), the LexUR and ASF point estimates are close (0.268 and 0.266). Bars show replication-level paired-bootstrap 95% intervals for both grouped and tail loss; no equivalence claim is inferred from interval overlap.

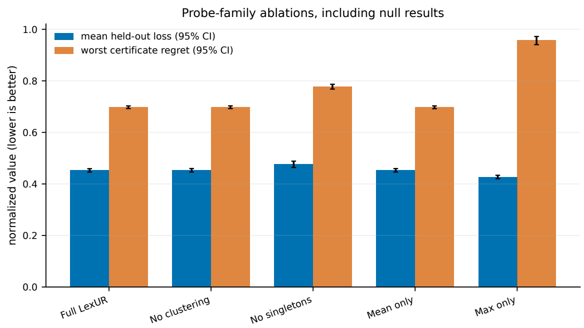

Figure 12: Ablation (concave, $m = 8$). Removing the singletons hurts both loss and certificate regret. No clustering and mean-only coincide with full LexUR in this independent-criterion setting, while a max-only family lowers mean loss but inflates worst-case certificate regret. Bars show 95% bootstrap intervals.

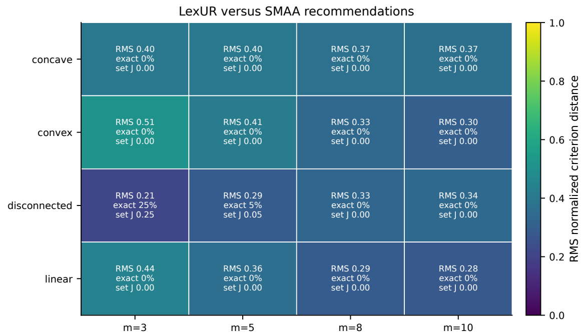

Figure 13: LexUR versus the SMAA first-rank recommender. Cells report exact agreement, tolerance-set Jaccard overlap at $\tau = 10^{-4}$, and RMS normalised criterion distance $\sqrt{m^{-1}\sum_i (r_i^{\mathrm{LexUR}} - r_i^{\mathrm{SMAA}})^2} \in [0, 1]$, which is comparable across m.

comparable to simple scalarizations (TOPSIS, CP) and roughly two orders of magnitude faster than evaluating robust Monte-Carlo baselines (SMAA, MMR) with 1,000 weight samples.

7.9. Stochastic extension

Table 8 summarises the different stochastic metrics used to validate the extensions.

We validate the sample-size recovery (Proposition 2) directly. On noisy concave instances ($m = 6$) we draw n observations per criterion and form the upper-confidence objectives of Eq. (9) (confidence-aware LexUR, $\gamma = 0.95$), then apply deterministic LexUR; Figure 15 reports how often the resulting choice matches the noise-free LexUR winner. Recovery rises from 0.055 at $n = 1$ to 0.84 at $n = 100$, consistent with the consistency statement. At the declared $\tau = 10^{-4}$ the noise-free tolerance classes are singletons in this experiment, so exact recovery and tolerance-class coverage coincide.

7.10. Smart-grid dispatch case study

We apply LexUR to a six-objective stochastic 24-hour economic dispatch formulated on a synthetic 14-generator dispatch instance constructed using the IEEE 14-bus demand/network template. We emphasise that this is an *aggregate* single-bus dispatch with a lumped loss term, not a network-constrained AC/DC optimal power flow; the IEEE 14-bus data is used only as a demand/generator template. The objectives track total generation cost, CO_2 emissions, unreliability (loss of load probability), inter-hour ramping

Table 7: Parameter diagnostics. Left: Held-out loss across clustering thresholds θ. Right: Paired nadir perturbations reuse the same candidate problems and held-out utilities at every realised RMS error level; quality changes remain small while recommendation identity changes frequently.

θ	linear	concave	convex	discon.
0.3	0.179	0.459	0.036	0.351
0.5	0.179	0.452	0.036	0.355
0.8	0.176	0.452	0.037	0.360

nadir err.	loss	flip rate
0%	0.451	0%
5%	0.453	23%
10%	0.457	50%
20%	0.457	73%
30%	0.458	90%
50%	0.446	97%

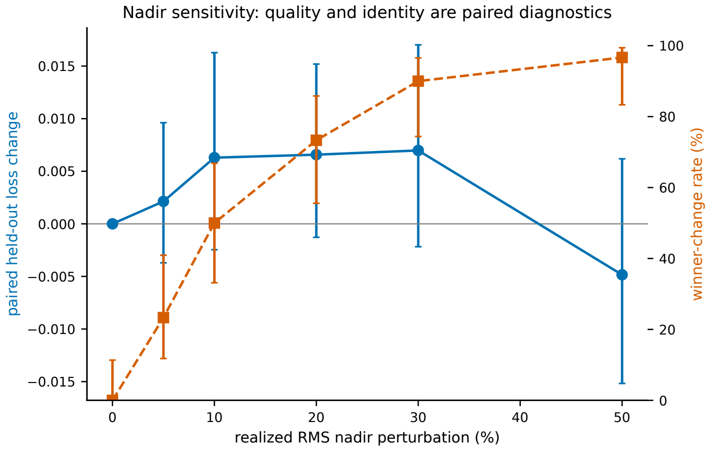

Figure 14: Nadir sensitivity. As the nadir estimate is corrupted, held-out *quality* change remains small while winner identity changes frequently. Each replication reuses one candidate set, held-out utility draw, and perturbation direction across all realised RMS error levels. Points show paired means with 95% bootstrap intervals (quality) and Wilson intervals (winner-change rate).

wear, non-renewable generation fraction, and the peak-to-average demand gap. The system is evaluated under Monte-Carlo renewable (wind/solar) generation scenarios. The 163 non-dominated dispatch policies were generated *a priori* by sampling a population of 250 candidate dispatch policies (Dirichlet-weighted generator allocations) under the renewable scenarios and retaining the non-dominated set; this illustrates how LexUR interfaces with *any* established *a priori* multi-objective generation method (here a simple population sampler, but an evolutionary algorithm such as NSGA-II would serve identically) as a post-hoc robust selector without requiring LexUR-specific candidate generation. The full dispatch model (demand profile, generator limits, ramping, scenario generation) is formulated in Appendix B and fully specified in the released code. Relative to the distance/averaging methods, LexUR accepts a negligible increase in *mean* held-out loss (0.103 vs 0.100) for a 23% reduction in *worst-case* loss (0.282 vs 0.366; knee 0.623). Crucially it *explains* the choice: the certificate (Figure 16) flags the peak-shaving gap (f_6, regret 0.40) and the non-renewable fraction (f_5, 0.40), then the ramping/peak coalition $\max(f_4, f_6)$ (0.36), telling the DM not only which dispatch to adopt but where it remains fragile. MMR selects the same dispatch in this synthetic case; LexUR's additional output is the complete, semantically labelled probe certificate rather than a distinct point choice.

8. Discussion and limitations

What LexUR is, and is not.. LexUR is a robust, certifiable decision rule. LexUR is competitive with the strongest robust scalarisations on tail held-out loss—non-inferior to achievement scalarisation and sampled linear minimax regret (negligible effect sizes)—while on average-case loss the classical averaging methods do better. LexUR's distinctive value is the *combination* it offers that no single competitor

Table 8: Summary of stochastic metrics reported across Sections 7.8 and 7.9.

Stochastic experiment	Metric	Compared methods	Result & Interpretation
Sample-size recovery	probability of matching noise-free LexUR winner	stochastic LexUR only	recovery rises with S; consistency diagnostic
Deterministic vs stochastic LexUR	tail held-out loss	det. LexUR vs stoch. LexUR	0.383 vs 0.383; stochastic adjustment does not hurt tail loss
Out-of-sample risk	worst-criterion risk	stoch. LexUR vs TOPSIS/SMAA	near 0 vs 0.28/0.52; confidence-aware certificate reduces risk

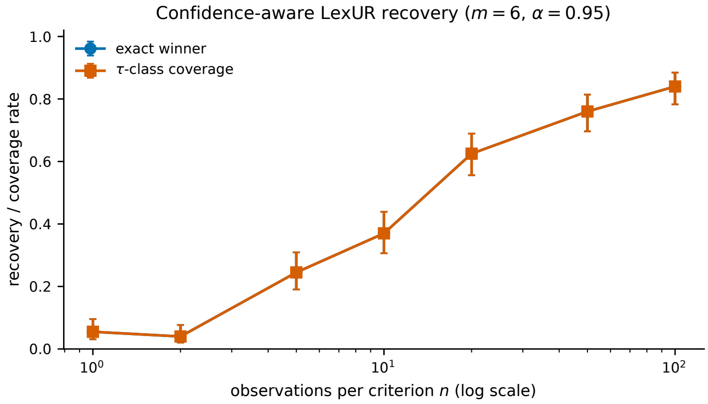

Figure 15: Confidence-aware LexUR recovery with $\alpha = 0.95$ under Gaussian noise of scale 0.15 times each criterion's cross-candidate standard deviation (concave, $m = 6$). Exact-winner recovery and coverage of the noise-free $\tau = 10^{-4}$ class use 40 problems and five resamples per sample size; bars are Wilson 95% intervals, and the observation-count axis is logarithmic.

matches: worst-case robustness competitive with the strongest robust scalarisations across additive *and* non-additive preferences (within one Nemenyi CD of MMR, VIKOR and ASF, though behind the single-point hypervolume score on average rank), reduced sensitivity to redundant criteria, an interpretable certificate, and natural stochastic and multi-stakeholder extensions (the latter currently exploratory, Appendix C). Where a DM cares only about average-case linear performance, a weighted sum or SMAA is appropriate; where the DM cannot state preferences and must defend a single robust, explainable choice, LexUR is designed for exactly that situation. Table 9 provides a concise summary mapping the central claims of the paper to their theoretical or empirical evidence.

Table 9: Claim-evidence matrix mapping theoretical properties and empirical findings to their supporting evidence.

Claim	Evidence
Pareto compatibility	Theorem 2 (general) and Theorem 4 (non-negative probes)
Existence and completeness	Theorem 1
Tolerance-rule stability	Theorem 3 (margin condition)
Reduced redundancy bias	Table 6 (grouped loss)
Robustness on tail held-out loss	Figure 9 and Table 4
Stochastic consistency	Proposition 2 and Figure 15
Smart-grid applicability	Appendix B and Figure 16

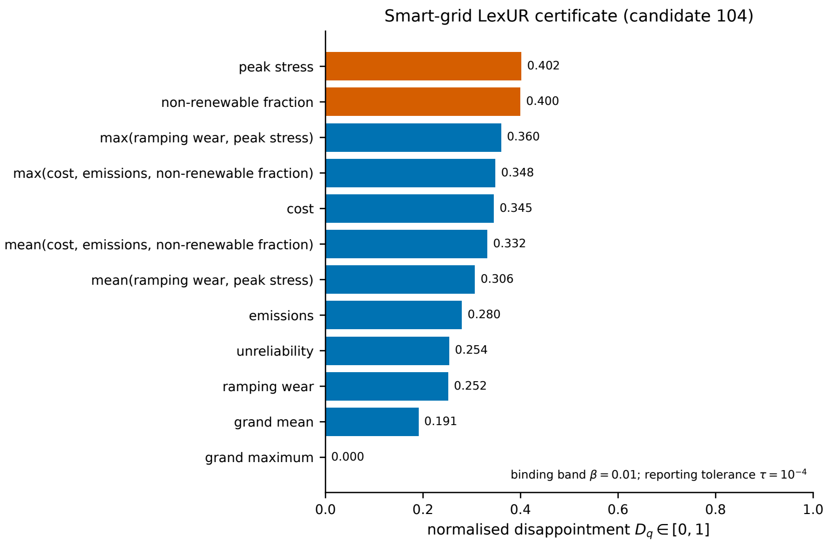

Figure 16: Complete LexUR regret certificate for smart-grid candidate 104. All 12 probes are shown on the common $[0, 1]$ scale with semantic labels. Orange denotes the presentation-only binding band $\beta = 0.01$ from the maximum; the numerical reporting tolerance is $\tau = 10^{-4}$. MMR selects the same candidate in this synthetic case.

Relationship to existing rules.. LexUR generalises rather than replaces. A single grand-maximum probe recovers the Chebyshev/ASF rule; a single weighted-mean probe recovers a weighted sum, and the grand mean recovers its equal-weight case; the singleton family recovers lexicographic maximin of normalised regret. The empirical question is whether the richer adaptive family helps, and the answer is that it does on redundant and asymmetric problems while not materially hurting on the tested standard geometries.

Independence of Irrelevant Alternatives (IIA) and Set-Dependence.. Because both the normalisation bounds $(z^\star, z^{nad})$ and the probe anchors $(q^\star, q^w)$ are computed empirically over the active candidate set A, LexUR does not satisfy Independence of Irrelevant Alternatives (IIA). Adding or removing a non-selected, non-dominated alternative that defines an extremum shifts the normalisation for *all* alternatives, which can change the sorted disappointment vectors and cause rank reversal. This is structurally the same vulnerability identified in TOPSIS (§1). This set-dependence explains the high winner-flip rates observed in our nadir-sensitivity experiments (Table 3). We must therefore unequivocally state: unless normalisation bounds are fixed *a priori* by the decision maker, point-estimate normalisation should be avoided in favour of the interval/stability analysis (Option B) defined in §3. Deploying LexUR with conservative interval bounds is mandatory when stability against candidate-set perturbations is required.

The Certificate's Decision Value.. While robust scalarisations like ASF provide similar worst-case guarantees, LexUR's distinct value lies in its auditable certificate. A standard sensitivity sweep over ASF reference points or a SMAA acceptability index reveals *how often* an alternative wins, but does not explain *why* it loses in specific contexts. In contrast, the LexUR certificate explicitly isolates the binding constraints: it names the specific declared probe (e.g., "the average of cost and emissions") that forces the maximum disappointment, giving the decision maker a direct semantic justification for the compromise that opaque distributional indices omit. Figure 17 shows the anatomy of such a certificate.

Scope, Parameters, and Limitations.. LexUR is a lexicographic minimax regret selector. It complements exploratory tools like SMAA by providing a single, auditable point recommendation when the DM must defend a robust choice. The framework relies on several explicit choices, summarised in Table 10. LexUR's contribution is that these choices are declared and separated from the final recommendation.

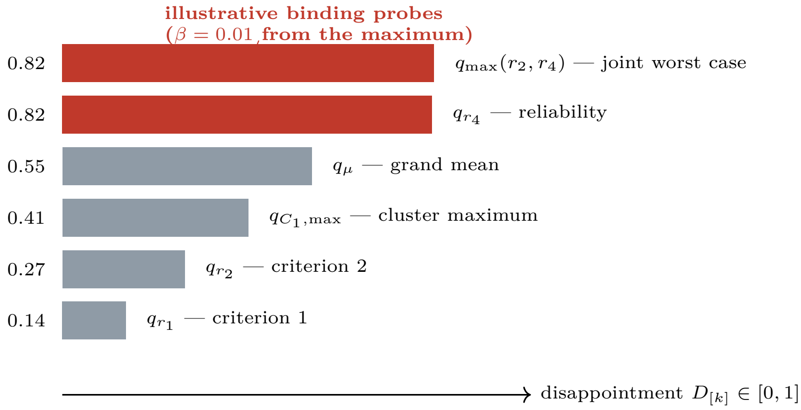

Figure 17: Anatomy of an illustrative regret certificate. Probe disappointments are sorted worst-first while retaining their semantic labels. The highlighted probes lie within the presentation band $\beta = 0.01$ of the maximum and are called binding; lower entries are secondary lexicographic constraints. The schematic values are not data from the smart-grid case.

Table 10: Consolidated list of LexUR parameters, their roles, and recommended defaults.

Param	Role	Default	Note
$z^\star, z^{nad}$	Normalisation bounds	Empirical over A	Must use fixed *a priori* bounds if IIA/stability is required.
$\mathcal{Q}$	Probe family	Adaptive family	Defines the semantic interpretations of preference.
θ	Clustering threshold	0.6	Empirically insensitive over $[0.4, 0.8]$.
τ	Leximax tolerance	10^{-4}	Set to the scale of the disappointment estimation error.
ε_q	Degeneracy / range floor	10^{-6}	Minimum admissible probe/criterion active range; below it the probe/criterion is flagged degenerate. The empirical stability margin Δ of Theorem 3 is an instance property, not a tunable parameter.
α	Stochastic confidence	0.95	Used only in the stochastic extension (§6).

Threats to validity.. The held-out families, though disjoint from the probes, are themselves a modelling choice; we mitigate this by spanning linear, Chebyshev, ASF and CES forms and by reporting per-family results. Candidate sets are synthetic but cover eight audit geometries, with four canonical geometries used in the mechanistic study, and the dispatch illustration is also synthetic; extending to a wider library of real MCDA datasets is future work.

Interactive elicitation using certificates.. The certificate could support a future interactive workflow in which the DM reviews the binding probes and revises the declared family. For example, a DM might relax a cost-focused probe after seeing its effect on the recommendation, after which LexUR can be recomputed. This possibility has not been evaluated here: removing a separating singleton can weaken Pareto-exclusion guarantees, so an interactive protocol would require explicit safeguards and user validation.

Future work.. Natural next steps include large-scale empirical validation of the continuous direct formulations (§4) on high-dimensional problems; and full development and evaluation of the exploratory Rawlsian multi-stakeholder variant. Finally, extending the family to include non-linear probes would significantly broaden LexUR's expressive power. For instance, capturing Cobb-Douglas structures (constant elasticity of substitution) could be achieved by evaluating linear probes in log-transformed criterion space, while modelling threshold-based satisficing preferences with interaction terms would require explicitly incorporating boolean conditionals into the anchor computations.

9. Conclusion

We have presented the Leximax Universal-Regret core, a Pareto-compatible decision rule that replaces the efficient set as the final object of multicriteria analysis with an auditable certificate. LexUR operates as a lexicographic minimax regret selection rule: it ranks alternatives by the lexicographically sorted vector of normalised achievement shortfalls over a declared family of monotone probes, and returns a robust stability class (often a singleton recommendation) together with an explanation of where it remains fragile. We proved existence and completeness, Pareto compatibility *under a separation condition*, dominated-solution exclusion, stability of the tolerance-based rule under bounded perturbations, and a disappointment-perturbation bound with explicit constants, and we gave a redundancy-aware, correlation-clustering probe construction that keeps the family linear in the number of criteria. A replicated held-out study over ten preference families and a stochastic smart-grid case showed that LexUR attains worst-case robustness competitive with the strongest robust scalarisations—performing non-inferiorly to achievement scalarisation and sampled minimax regret, while a single-point hypervolume score (HV_1) obtains the best average rank—and does so while reducing sensitivity to redundant criteria, certifying its choices, and extending naturally to noisy criteria.

The central message is that robust, explainable selection can be obtained from a declared family of monotone questions rather than from elicited scalarisation weights. In structured cases (e.g. linear MOPs) LexUR can sometimes be computed directly, without front enumeration; in the general case, however, it remains primarily a *post hoc* candidate-set selector, and we make no claim that it replaces front generation universally. This robustness also relies entirely on the reliability of the supplied normalisation bounds; if true nadirs are uncertain, the framework must be deployed conservatively using interval estimates. By relocating preference choices into a declared class of questions, LexUR turns a candidate set into a defensible, explainable recommendation—a low-disappointment compromise across the declared interpretations of the criteria, accompanied by a certificate that reports which questions remain most binding.

Reproducibility.. All code, generated data, and scripts that regenerate every table and figure in this paper are released in an open-source repository (URL withheld to preserve anonymity during review; it will be provided in the camera-ready version), including the four-geometry benchmark, the redundancy benchmark, the statistical tests, the sensitivity analyses, and the smart-grid case study.

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Appendix A. Held-out preference families and broadened-protocol specification

This appendix gives the formal specification of the held-out evaluation families and the broadened audit protocol of §7.2, for reproducibility. None of these families is used to construct LexUR probes; they exist only to score recommendations out of class.

Notation and scoring.. Let $r(x) \in [0,1]^m$ be the active-set normalised criterion vector of alternative x (minimisation, lower is better). Each family produces, for T sampled parameterisations, a utility $u(\cdot)$ (higher is better); a recommendation is scored by the held-out loss of Eq. (10), $L_u(x) = \big(u(x_u^\star) - u(x)\big)/\big(u(x_u^\star) - u(x_u^-)\big)$, with $x_u^\star, x_u^-$ the in-sample best and worst alternatives for u on A. Unless stated otherwise, importance weights are drawn from a Dirichlet distribution $w \sim \mathrm{Dir}(\alpha\mathbf{1})$, whose shape parameter $\alpha \in \{0.2, 1, 5\}$ yields sparse/extreme, uniform-simplex, and balanced preferences respectively.

The ten families.. Writing $u(x)$ for the utility (higher better), the families are:

- **linear:** $u(x) = -w^\top r(x)$, $w \sim \mathrm{Dir}(\alpha\mathbf{1})$.
- **sparse-linear:** as linear but only the $k = \lceil m/3 \rceil$ largest weights are kept and renormalised to sum to one (a DM attending to a few criteria).
- **Chebyshev:** $u(x) = -\max_i w_i r_i(x)$ (weighted Chebyshev / pure ASF direction).
- **augmented ASF:** $u(x) = -\big(\max_i w_i r_i(x) + 0.05\, w^\top r(x)\big)$, i.e. the Chebyshev term plus a small $\rho = 0.05$ linear augmentation on the same weights.
- **L_p compromise:** $u(x) = -\big(\sum_i w_i r_i(x)^p\big)^{1/p}$ with exponent $p \sim \mathrm{Unif}(1.5, 4.0)$ drawn per utility.
- **CES:** $u(x) = -\big(\sum_i w_i r_i(x)^\varrho\big)^{1/\varrho}$ with elasticity exponent $\varrho \sim \mathrm{Unif}(2.0, 5.0)$ drawn per utility (constant-elasticity-of-substitution disutility). We note plainly that the L_p and CES families are *intentionally adjacent* rather than independent: both are norm-like disutilities of the same functional form, differing only in the exponent range. Likewise Chebyshev and augmented ASF are

near-identical by construction (§7). We therefore do not claim ten *independent* preference geometries; the families are chosen to span a graded distance from the probe family (in-class to adversarial), not to be mutually orthogonal.

- **OWA:** $u(x) = -\sum_{k=1}^{m} \omega_k \, r_{[k]}(x)$, where $r_{[k]}$ is the k-th largest normalised criterion and the ordered weights $\omega = \mathrm{sort}^{\downarrow}\big(\mathrm{Dir}(\mathbf{1})\big)$ are a descending-sorted uniform-simplex draw (emphasis on worst coordinates).
- **Choquet-inspired (2-additive, max-interaction):** $u(x) = -\big(a^\top r(x) + \sum_{i<j} b_{ij} \max(r_i(x), r_j(x))\big)$ with non-negative masses $a_i \sim \mathrm{Unif}(0,1)$ on singletons and $b_{ij} \sim \mathrm{Unif}(0,0.5)$ on pairs. This is a monotone non-additive pairwise-interaction utility in the *spirit* of a 2-additive Choquet integral; we deliberately avoid claiming it is exactly the standard 2-additive Choquet integral (whose Möbius/capacity form involves min terms for the pairwise interactions), and use it only as a genuinely non-additive held-out family.
- **satisficing:** $u(x) = -\sum_i \max\big(r_i(x) - \vartheta_i, 0\big)$ with per-criterion aspiration thresholds $\vartheta_i \sim \mathrm{Unif}(0.2, 0.6)$, penalising only exceedances of the aspiration levels.
- **max-min (group regret):** three stakeholder weight vectors $w^{(1)}, w^{(2)}, w^{(3)} \sim \mathrm{Dir}(\alpha\mathbf{1})$ define $u(x) = -\max_s (w^{(s)})^\top r(x)$, the worst-stakeholder linear disutility.

Families are grouped by distance from the probe family as *in-class* (linear, sparse-linear, Chebyshev, augmented ASF), *adjacent* (L_p, CES, OWA), *out-of-class* (Choquet, satisficing) and *adversarial* (max-min group regret); this scoping is fixed in the frozen config.

Protocol grid.. The broadened protocol of §7.2 sweeps candidate-set sizes $N \in \{100, 300, 1000\}$, criteria $m \in \{3, 5, 8, 10, 15, 20\}$, the eight geometries (linear, convex, concave, disconnected, degenerate, asymmetric, many-knee, irregular), and 50 replications, for $3 \times 6 \times 8 \times 50 = 7{,}200$ paired instances. A Dirichlet shape $\alpha \in \{0.2, 1, 5\}$ is assigned to each replication by deterministic cycling ($\alpha = \alpha_{\mathrm{rep} \bmod 3}$), so the three shapes are nearly balanced $(17, 17, 16$ over the 50 replications) within each $(N, m, \text{geometry})$ cell. The paired non-inferiority confidence intervals (§7.2) use a cluster bootstrap with the instance keys as resampling units. The same candidate sets and held-out utility draws are shared across all methods within an instance to ensure paired comparisons. All parameter distributions above, the random seeds, and the frozen acceptance gates are fixed in the released configuration file and reproduced by a single command.

Front geometries.. Let s denote a point drawn on the unit simplex (symmetric Dirichlet) and d a vector with $d_i = \sqrt{U_i}$, $U_i \sim \mathrm{Unif}(0,1)$; every set is then reduced to its non-dominated subset. The eight geometries are:

- **linear:** $F = 0.5\, s$ (hyperplane $\sum_i f_i = 0.5$).
- **convex:** $F = s^{\circ 2}$ (componentwise square; convex front $\sum_i \sqrt{f_i} = 1$).
- **concave:** $F = d/\|d\|_2$ (concave spherical front $\sum_i f_i^2 = 1$).
- **disconnected:** the concave sphere with the first coordinate offset by $0.6\,k$, $k \in \{0, 1, 2, 3\}$, opening gaps between patches.
- **degenerate:** the $s^{\circ 2}$ front with the last criterion tied to the first ($f_m \leftarrow f_1$), giving a lower-dimensional front.

Table A.11: Baseline selection rules used in §7. All scores are on the normalised matrix $r \in [0,1]^m$ (lower r better). Unless noted, ties are broken by the first index attaining the optimum (the default of `argmin`/`argmax`).

Method	Score	Select	Parameters / tie-break
TOPSIS	relative closeness $d^-/(d^- + d^+)$ to ideal/anti-ideal in weighted space	arg max	$w = \mathbf{1}/m$; first index
CP	L_p distance $(\sum_i w_i r_i^p)^{1/p}$ to ideal	arg min	$w = \mathbf{1}/m$, $p = 2$; first index
VIKOR	$Q = v\frac{S-S^-}{S^+-S^-} + (1-v)\frac{R-R^-}{R^+-R^-}$, $S = \sum_i w_i r_i$, $R = \max_i w_i r_i$	arg min	$w = \mathbf{1}/m$, $v = 0.5$; first index
Knee	distance below the hyperplane $a^\top r = 1$ through the m per-criterion extremes (where a solves $Ea = \mathbf{1}$, E the matrix of extreme rows), $1 - a^\top r$	arg max	first index; CP fall-back if E singular
HV_1	single-point hypervolume (box volume) $\prod_i \max(1-r_i, 0)$	arg max	ref. $\mathbf{1}$; first index
RW	frequency of being $\arg\min_x w^\top r$ over random weights	arg max count	200 weights $\sim$ Dir($\mathbf{1}$); first index
ASF	augmented Chebyshev $\max_i r_i/w_i + \rho\sum_i r_i$, reference = ideal	arg min	$w = \mathbf{1}/m$, $\rho = 10^{-4}$; first index
SMAA	first-rank acceptability under linear value $-w^\top r$	arg max	1000 weights; ties by expected value
MMR	$\max_w \big(w^\top r(x) - \min_{y\in A} w^\top r(y)\big)$ (Savage regret)	arg min	1000 weights $\sim$ Dir($\mathbf{1}$); first index
ChebMMR	$\max_w \big(\max_i w_i r_i(x) - \min_{y\in A} \max_i w_i r_i(y)\big)$	arg min	1000 weights $\sim$ Dir($\mathbf{1}$); first index
LexUR	leximax of the sorted normalised probe disappointments	lex. min	adaptive probes, $\theta = 0.6$, $\tau = 10^{-4}$; tolerance set

- **asymmetric:** the concave sphere with criteria scaled by linspace$(1, 5, m)$ (heterogeneous criterion ranges).
- **many-knee:** the concave sphere warped by $F \mapsto F + 0.15\sin(6F)$ to create multiple knees, then clipped to $F \geq 0$.
- **irregular:** the concave sphere with multiplicative noise $F \mapsto F \odot (1 + 0.1\,\mathcal{N}(0,1))$, then clipped to $F \geq 10^{-3}$ so all criterion values stay non-negative.

All eight constructions yield non-negative criterion values (by clipping where noted), so the subsequent active-set normalisation maps them into $[0,1]^m$.

Baseline specification.. Table A.11 compactly specifies every comparator, all operating on the active-set normalised matrix $r \in [0,1]^m$ (minimisation). w denotes equal weights $\mathbf{1}/m$ unless a method samples weights.

Appendix B. Smart-grid economic dispatch formulation

The case study in §7 evaluates a 24-hour multi-objective economic dispatch on a synthetic 14-generator instance constructed using the IEEE 14-bus demand/generator template. To prevent any misreading of the model's fidelity, we describe the actual implementation precisely; it is deliberately a lightweight *priority-dispatch heuristic*, *not* a binary unit-commitment MILP and *not* a network-constrained AC/DC optimal power flow. It is an *aggregate* (single-bus) dispatch: the IEEE 14-bus data supplies the demand profile and the generator cost/emission/ramp template only, and no nodal power-flow, transmission-limit, line-loss, or binary commitment constraints are modelled (the dispatch balances the hourly demand $d_t(\omega)$ directly).

Candidate policy and dispatch rule.. A candidate policy is a non-negative *dispatch-priority* vector $\pi \in \mathbb{R}^{14}_{\geq 0}$, normalised to $\sum_g \pi_g = 1$; there are no binary commitment variables, and all generators are treated as available (renewable units have hour- and scenario-dependent available capacity). For each hour t and scenario ω, let $C_{gt}(\omega)$ be the available capacity ($C_{gt} = P_g^{\max}$ for thermal units, and the renewable cap scaled by the realised solar/wind availability for renewable units). The dispatch is computed by a two-pass priority rule:

$$p_{gt}(\omega) = \min\bigl(\pi_g \, d_t(\omega),\, C_{gt}(\omega)\bigr) \text{ (nominal share, capped)}, \tag{B.1}$$

after which any residual demand $d_t(\omega) - \sum_g p_{gt}(\omega)$ is filled greedily in proportion to the remaining headroom $C_{gt}(\omega) - p_{gt}(\omega)$, up to capacity. No ramp constraint is enforced; inter-hour ramping is *measured* as the wear objective below, not constrained. Renewable energy beyond what is dispatched is simply not used (there is no explicit curtailment variable). Any demand that still cannot be met is recorded as load shedding,

$$s_t^{\text{shed}}(\omega) = \max\Bigl(d_t(\omega) - \textstyle\sum_g p_{gt}(\omega),\, 0\Bigr), \tag{B.2}$$

which feeds the unreliability objective. Every sampled policy is therefore *feasible by construction*: capacity violations are impossible (dispatch is capped) and any supply shortfall is absorbed as load shedding rather than discarded or repaired.

The six objectives to be minimised are given below, each up to a fixed positive scaling constant that does not affect ranking or active-set normalisation. Let G_{th} and G_{ren} denote the thermal and renewable generator sets, c_g, e_g the fuel-cost and emission coefficients, and R_g the ramp scale.

1. **Cost:** $f_1(x) = \mathbb{E}_\omega\Bigl[\sum_t \sum_g c_g \, p_{gt}(\omega)\Bigr]$.
2. **Emissions:** $f_2(x) = \mathbb{E}_\omega\Bigl[\sum_t \sum_g e_g \, p_{gt}(\omega)\Bigr]$.
3. **Unreliability:** Loss-of-load probability, the fraction of scenarios in which any load is shed,

$$f_3(x) = \frac{1}{|\Omega|} \sum_{\omega \in \Omega} \mathbf{1}\left[\sum_t s_t^{\text{shed}}(\omega) > 0\right].$$

4. **Ramping wear:** expected ramp magnitude normalised by the per-generator ramp scale R_g (a wear normaliser, *not* an enforced ramp limit),

$$f_4(x) = \mathbb{E}_\omega\left[\sum_g \sum_{t>1} \frac{|p_{g,t}(\omega) - p_{g,t-1}(\omega)|}{R_g}\right].$$

5. **Non-renewable fraction:** expectation *of the per-scenario ratio* of thermal to total dispatched

energy (an expectation of a ratio, not a ratio of expectations),

$$f_5(x) = \mathbb{E}_\omega \left[\frac{\sum_t \sum_{g \in G_{\text{th}}} p_{gt}(\omega)}{\sum_t \sum_g p_{gt}(\omega) + \varepsilon} \right].$$

6. **Peak stress:** a peak-shaving proxy equal to the peak demand scaled by the priority weight *not* pre-allocated to thermal generators,

$$f_6(x) = d^{\text{peak}} \max\Big(1 - \textstyle\sum_{g \in G_{\text{th}}} \pi_g,\, 0\Big).$$

Each candidate policy is evaluated by Monte-Carlo sample-average approximation over 20 renewable scenarios drawn from the scenario model. The candidate set is generated *a priori*, independently of LexUR: we sample a population of 250 dispatch policies as Dirichlet-weighted generator allocations ($\text{Dir}(0.6 \cdot \mathbf{1})$ over the 14 generators, seed 2024), evaluate each under the scenarios, and retain the 163 non-dominated policies as the efficient candidate set. No evolutionary operators (selection, crossover, mutation) are used; a population metaheuristic such as NSGA-II could be substituted without changing LexUR's role as a post-hoc selector. All parameter values (generator data, costs, emissions, ramp-wear coefficients R_g, and scenario bounds), random seeds, and the exact scenario count are fixed in the open-source repository. This formulation serves as an illustrative application of the robust certificate framework.

Appendix C. Multi-stakeholder (Rawlsian) exploratory variant

Let $\mathcal{U} = \{1, \ldots, U\}$ index stakeholders, each with importance weights $w_u \in \Delta^m$ inducing a stakeholder-specific disappointment $D_q^u(x)$. An exploratory *Rawlsian LexUR* applies the leximax order to the worst-stakeholder disappointments, $\max_u D_q^u(x)$. It maintains natural fairness properties such as anonymity and worst-stakeholder protection. A full treatment of this extension is beyond the scope of this paper, but we include this preliminary formulation to demonstrate the flexibility of the regret-certificate framework.